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Exploring ethical values in software systems: A systematic literature review ☆

Razieh Alidoosti [a,b,*], Patricia Lago [a,**], Maryam Razavian [c], Antony Tang [a,d]

[a] Department of Computer Science, Vrije Universiteit Amsterdam, Amsterdam, The Netherlands
[b] Department of Computer Science, Gran Sasso Science Institute, L'Aquila, Italy
[c] Department of Industrial Engineering and Innovation Sciences, Eindhoven University of Technology, Eindhoven, The Netherlands
[d] Department of Computer Science and Software Engineering, Swinburne University of Technology, Melbourne, Australia

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ABSTRACT

Context: Ethics has attracted considerable attention in the field of software engineering. The prevalence of software-intensive systems and how they become an integral part of people's lives makes it essential for software engineers to care about the potential impacts and injustice of software systems on people. The social and ethical implications on individuals and society play an important role in the study of software engineering ethics.
Objective: The general aim of this paper is to study the state of the art in ethics in software engineering and to understand ethical considerations in software design and development.
Method: We conducted a systematic literature review (SLR) with an initial sample of 623 articles, of which 85 articles were selected as primary studies.
Result: We created a body of knowledge on stakeholders and software engineering ethical values. More specifically, we provided a stakeholders map to identify different types of system stakeholders and presented a value model to categorize ethical values and recognize value relations.
Conclusion: This SLR sheds light on the state of software engineering ethics. It highlights challenges such as the identification of indirect stakeholders, the lack of attention to their concerns, and the need for a stakeholder classification. The review emphasizes the importance of considering stakeholders' concerns to uncover ethical values and discusses methods for value extraction, with a focus on the value model. There are needs to address particular ethical values, resolve value conflicts, and operationalize ethical values in software design. These findings offer valuable insights for future research and practice, encouraging a comprehensive integration of ethical aspects into software engineering.

1. Introduction

Today, the study of ethics can be found in a variety of fields like medicine and business management (Chung and Khan, 2008); ethics as a branch of philosophy resolves the questions related to human value and morality by defining concepts like right[1]/good[2] and wrong[3]/bad[4] (Capurro, 1988; Yücel et al., 2009; Paul and Elder, 2013; Veatch, 1977). Over the last decade, ethics has received the attention of the software engineering field because of the high dependency of people's lives on software-intensive systems and the related social and ethical implications on individuals and society. In this context, ethics refers to a set of principles and rules providing guidance for software professionals (Gotterbarn, 2002).

Software-intensive systems, or *systems* for short, offer significant benefits to individuals and society by enhancing productivity, facilitating communication, and driving innovation across various sectors (Brynjolfsson and McAfee, 2014). However, these systems can also potentially harm individuals and society They can lead to financial loss, damage reputations, and, in some cases, even threaten human life.

Consider Therac-25, a computer-controlled radiation therapy system (Israelski and Muto, 2004; Huff and Brown, 2004). Therac-25 was

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* Correspondence to: Vrije Universiteit Amsterdam, De Boelelaan 1111, 1081 HV Amsterdam, The Netherlands.
** Corresponding author.
E-mail addresses: razieh.alidoosti@gssi.it (R. Alidoosti), p.lago@vu.nl (P. Lago).

[1] It is derived from "rectus" and means straight or according to rule.
[2] It implies something desirable and acceptable.
[3] It means not in accordance with what is morally right or good.
[4] It is the opposite of good and implies something undesirable, unacceptable, and not pleasing in every manner.

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designed to provide radiation therapy services essential for treating cancer. However, due to safety violations, the system massively overdosed radiation to six patients in the late 1980s, resulting in three deaths and three severe injuries. Similarly, consider Knight Capital Group, a financial services firm (Saltapidas and Maghsood, 2018). In August 2012, an error in the deployment of an automated trading system caused the execution of unintended stock trades, leading to massive market disruption. Although no physical harm occurred, the firm lost $440 million in 45 minutes, nearly leading to bankruptcy. This incident indirectly harmed shareholders, employees, and market stability, highlighting the catastrophic financial and reputational damage resulting from inadequate testing and prioritization during development. Research revealed that these accidents occurred because: (i) the designers gave priority to the system users (*i.e.,* machine operators and medical physicists) as the main stakeholders and only focused on the usability of the system; usability was potentially in conflict with safety; however, this conflict was never understood and accounted for (Leveson et al., 1995); and (ii) while reliability and safety are conceptually different qualities, they were considered interchangeable (Leveson et al., 1995). This example illustrates the importance of ethical values as one of the fundamental design concerns Ozkaya (2019).

When addressing ethical considerations, there are issues that one needs to take into account: (i) ethics is context-dependent, (ii) ethics is not a clear matter of right or wrong, or good or bad, and (iii) ethical issues are not easy to recognize and deal with. Whilst ethical principles can be used as a guideline in software engineering, ethics cannot be achieved simply by following a set of rules and standards (*e.g.,* ACM/IEEE software engineering code of ethics (Gotterbarn et al., 1997)) because ethical aspects are not rules or requirements that can be easily tested. As such, software engineers do not have explicit and systematic approaches to deal with human values in software design and development. There is a need for ethical considerations and embedding ethical values (such as welfare and human rights) in the building of software systems, such as software design (Thomson and Schmoldt, 2001; Berenbach and Broy, 2009; Turilli, 2008). Identifying such considerations requires a continuous process of *analysis for social life, seeking complexities instead of clarity, collective responsibility instead of individual, and focusing on dialogue instead of solutions* (Rashid et al., 2015).

Technical and non-technical decisions that address ethical considerations can be complex and challenging in software design, such as trade-offs for satisfying constraints like clients' needs, environmental considerations, social effects, and many other difficult and intangible ethical issues. In order to address such complexities, software engineers need to analyze the ethical aspects of the systems (*e.g.,* factors related to humanity, society, and the environment) and, in particular, consider the implications of their decisions (Thomson and Schmoldt, 2001; Vallor and Narayanan, 2015; Berenbach and Broy, 2009). Understanding the ethical considerations of systems depends on knowing the systems and their stakeholders. In this regard, it is necessary to foster the habit of reflecting on relevant stakeholders, assessing what is at stake for each stakeholder during software design and development process (Vallor and Narayanan, 2015). By identifying stakeholders who affect or are affected by the systems, determining how stakeholders' rights and well-being will be impacted, and considering ethical values in design decisions, software engineers can identify potential ethical issues, better understand the ethical implications of their decisions, and find a way to balance these impacts effectively.

In the context of this Systematic Literature Review (SLR), the primary objective is to assess the current state of research in *ethics in software engineering* (SE ethics) and to identify gaps and emerging trends for future research. Despite extensive research in the field of ethics and technology, we find challenges persist in recognizing relevant stakeholders and their concerns, extracting ethical values, resolving value conflicts, and translating these values into software requirements.

This paper is structured as follows. Section 2 discusses background information, including fundamental concepts and the motivation for conducting a literature review on SE ethics. Section 3 outlines our research methodology, including the systematic literature review protocol (research goal and questions, search strategy, selection criteria, and data extraction and synthesis) and the research process. The results are presented in Section 4, then discussed in Section 5. Section 6 discusses the threats to validity, and Section 7 concludes the paper.

2. Background

In this section, we describe the change of focus in ethics and software, providing insights about their relations. We discuss the term "ethical value" from a philosophical standpoint and explain the significance of value classification. Given the lack of a comprehensive value classification in software engineering, we explore existing classifications in philosophy (like Schwartz's value structure) as a basis. Further, we illustrate the existing approaches accounting for values in the context of software systems.

2.1. Concepts and definitions

2.1.1. Ethics and software

Ethics in software refers to a set of principles and rules, helping software engineers adhere to the welfare, justice, and safety of users and society, and guiding software engineers in their activities and decisions during the software design and development process (Gotterbarn, 2002).

As stated by Albrechtslund (2007), until the twentieth century, ethics has been occupied with theories like *utilitarianism* (Mill, 2016) and *deontology* (Gaus, 2001). These theories have dominated the ethical debate within philosophy. Recently, however, ethical debates have included an application perspective and controversial problems concerning software systems (Albrechtslund, 2007). With emerging systems and technologies, ethicists changed their focus from the consequences of a system, to the shaping and designing of a system (Lago , 2023). By increasing the awareness of designers about the contexts in which a system is supposed to be used, developers can consider more about humanities (including aesthetics and ethics). The focus of ethics has shifted from theory-oriented to the process of design; similarly, software systems have shifted their focus from solely technology-driven to value sensitivity (Albrechtslund, 2007).

2.1.2. Ethical values

The term *value* refers to "interests, pleasures, likes, preferences, duties, moral obligations, desires, wants, goals, needs, aversions and attractions, and many other kinds of selective orientations" (Williams, 1979). There are different definitions for value; for example, a value is a belief based on which a person acts by preference (Fritzsche, 1991); a value is a prescriptive belief, and an *ethical* value is a prescriptive belief about what is "right" or "wrong" (Fritzsche, 1991); a value is a principle for action including abstract goals in life and modes of conduct that an individual or a group of people consider more suitable in a given context (Braithwaite, 1998); a value is not the same as ideal, norm, or espoused belief about the good, but is about operating criterion for action (Hutcheon, 1972). In this study, value refers to what a person or group of people consider important in life, like human welfare, ownership and property, privacy, trust, and autonomy.

Classification can aid in comprehending ethical values (Rescher, 1969). There are different value classifications that provide explicit categories of human values, such as *list of values* (Kahle et al., 1988), *personal value scale* (Wallace, 1966), and *Schwartz's value structure* (Schwartz, 1994). Classification approaches can vary in terms of their purposes and the principles of organizing values. Schwartz's value structure is one of the widely acknowledged tools for organizing values,

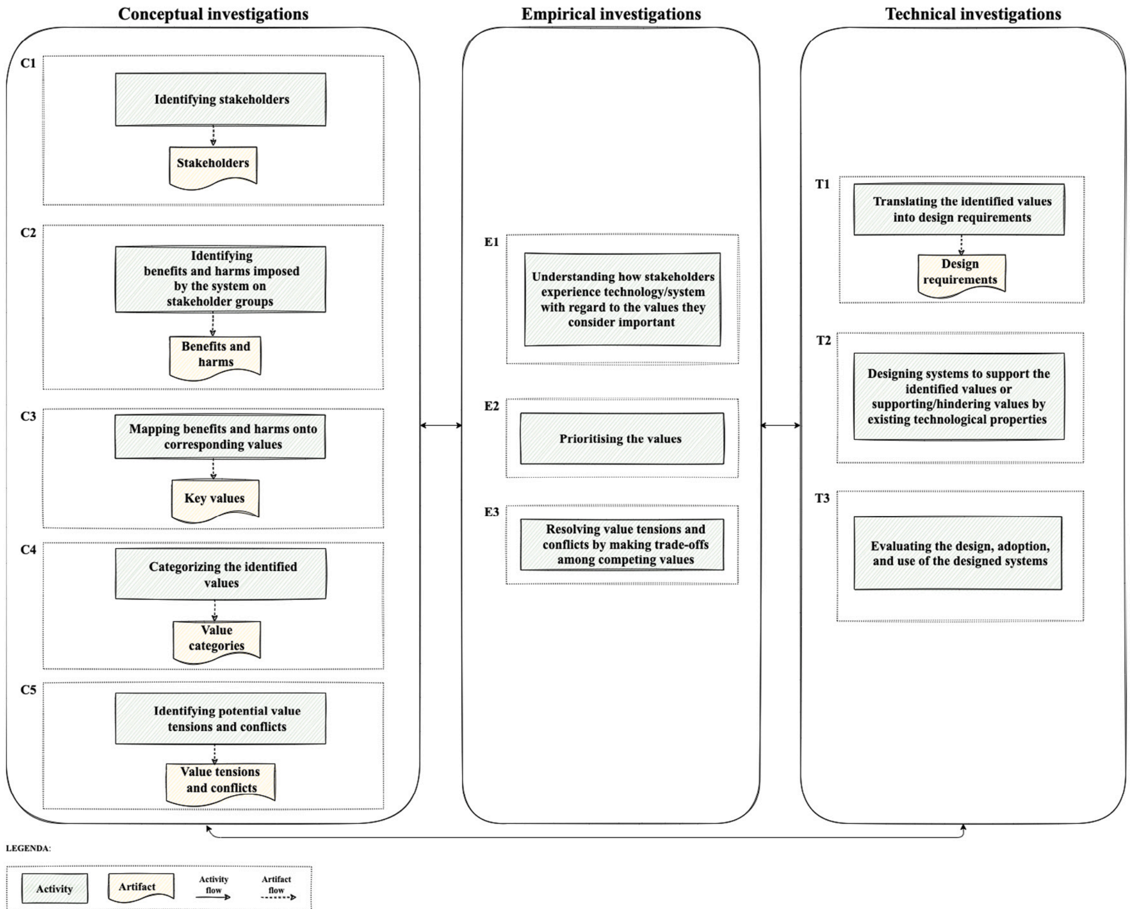

Fig. 1. An overview of the investigations and activities in VSD methods.

including 59 human values divided into 10 categories (each with different motivational goals) (Schwartz, 1992, 2012). This structure not only delineates the distinct value categories but also highlights the relations among them. These relations can be any kind of conflict or congruity. To better understand these relations, Schwartz describes them using two main dimensions: (i) self-enhancement vs. self-transcendence and (ii) openness to change vs. conservation. The first orthogonal dimension shows conflict among the values that focus on social superiority and esteem (*i.e.,* value categories of power and achievement) and the values that focus on the enhancement of others and the transcendence of selfish interests (universalism and benevolence). The second orthogonal dimension shows conflict among the values that focus on intrinsic interest in novelty and mastery (self-direction and stimulation) and those that focus on preserving existing social arrangements and harmony in relations (security, conformity, and tradition). The circular arrangement of values, defined by these two dimensions, establishes a motivation continuum. Values situated closer to each other in either direction around the circle share more similar motivations, indicating a higher likelihood of congruent relationships. Conversely, values positioned further away from each other tend to have contrasting motivations and are more prone to conflicting relationships.

The Schwartz's value structure provides a general, comprehensive, and organized structure that a number of studies have cited when discussing or classifying ethical values. The use of this structure is justified by its alignment with established theoretical concepts, as well as its extensive validation and refinement through empirical research in diverse cultures and contexts (Schwartz, 1992).

2.1.3. *Supporting values*

There exist various approaches to account for human values in the context of software systems. *Computer ethics* focuses on the analysis of the social impact of computer technology and the formulation of policies for the ethical use of technology (Moor, 1985); *Social Informatics* focuses on social aspects of computerization (Kling, 2000); *Computer Supported Cooperative Work (CSCW)* focuses on the design of technologies to help people do group work and have effective collaborations in their environment (Schmidt and Bannon, 1992); *Participatory Design* focuses on embedding democratic values into the practice (Carroll and Rosson, 2014); *Value Sensitive Design (VSD)* focuses on incorporating human values throughout the design process of technology (Winkler and Spiekermann, 2018; Manders-Huits, 2011; Davis and Nathan, 2015; Friedman et al., 2013).

Among these approaches, VSD has gained significant prominence in various domains, including software engineering. What makes VSD a popular choice in this domain is its holistic and integrated methodology. Unlike other approaches, VSD encompasses three primary investigations (see Fig. 1), each serving a specific purpose: (i) conceptual, (ii) empirical, and (iii) technical.

While VSD provides a structured framework for integrating values into design, it has some limitations. One challenge is addressing the variability of values across different cultures, as universal values need to be translated to practical impacts in specific contexts. Although heuristic value lists can assist designers, they may also introduce designer biases, there is a need to carefully consider both stakeholder and designer values. Additionally, VSD lacks concrete methods for empirical investigations, necessitating more specific techniques to capture lived experiences and empower stakeholders. The success of VSD often depends on the expertise of the design team, which may require social scientists or ethicists, and in industry settings these ethical expertise may be limited (Friedman et al., 2013).

Conceptual investigations provide philosophically-informed analyses of the central issues under investigation and focus on identifying values and stakeholders. This investigation focuses on questions like: What are values? Whose values should be considered in the design process? How can technological designs support or diminish values? How to conduct trade-offs among competing values (*e.g.,* anonymity vs. trust) in a design? Conceptual investigations are informed by empirical investigations of technology.

Empirical investigations have a role in the understanding and experiences of the people affected by technological designs. To do these kinds of investigations, the quantitative and qualitative methods in social science research are applicable. These investigations question the following: How are values perceived by stakeholders in a given context? How do stakeholders prioritize values in design trade-offs?

Technical investigations focus on the impacts of technology and implementation. They encompass two main activities: evaluating (i) how existing technologies support or undermine human values, and (ii) how creating and refining technologies align with human values.

VSD serves as a crucial theoretical framework for comprehending the activities related to integrating ethical values. The choice of this framework is motivated by the fact that all the primary studies we have reviewed rely on VSD as the foundational approach for addressing ethical values in software design. The use of this approach is warranted by its inherent characteristics, as it offers a systematic, interdisciplinary, and user-centered approach to addressing ethical considerations. By incorporating ethical considerations right from the inception of the design process, VSD facilitates the development of systems aligned with user values, helps mitigate ethical risks, and contributes to the responsible development of technology.

2.2. Need for a systematic review on SE ethics

Over the last decade, ethics has received increasing attention from the software engineering field and has opened a new research direction in this area, *i.e.,* SE ethics. This direction has led to a focus on the non-technical and ethical aspects of software systems, as well as the ethical issues caused by such systems. An example of ethical issues is using web browsers where end users aim to interact with web data while e-commerce vendors use personal data by exploiting tracking and profiling technologies. Doing so without explicit informed consent violates the privacy of end users and undermine the ethical values of privacy and confidentiality (Xu et al., 2012). Addressing such issues and supporting ethical values have mostly been neglected by previous research. From the literature, we found only one review study (Perera et al., 2020) exploring ethical aspects, especially human values in software engineering.

Perera et al. (2020) used the Schwartz's values structure and investigated the extent to which human values have been considered in SE research. They only found a small number of papers (*i.e.,* 216 out of 1350 papers) that investigated human values, and the values of security and privacy are the main focus of SE research. While this work aims to present the prevalence of ethical values among SE studies, it does not guide conceptualizing and classifying the values or recognizing the relations among values.

This review aims to help researchers gain an overview of SE ethics and a summary of the current evidence relevant to our research questions, from which we focus on (i) key stakeholders; (ii) ethical values; and (iii) main methods and approaches. They provide a window to help identify the issues and problems relevant to research on SE ethics and raise questions and directions for future research.

3. Systematic literature review methodology

In this section, we provide a detailed overview of our systematic literature review approach, following the established guidelines by Kitchenham (2004).

3.1. Research goal and questions

The objective of our SLR is to characterize research on ethics and ethical aspects in the design and development of software systems. This leads to the following research questions (RQs), each provided with the underlying rationale:

- **RQ1:** Who are the key stakeholders in software-intensive systems that affect or are affected by ethical values? **Rationale:** Identifying stakeholders is crucial because analyzing their concerns requires first understanding who they are. Stakeholder identification enables examination of how they are involved in or impacted by ethical values in software systems.
- **RQ2:** What are ethical values and the relations between them? How can ethical values be elicited from the relevant stakeholders? **Rationale:** Ethical values serve as a critical design aspiration and can be integrated into the software design and development process to address ethical considerations and resolve issues. These values can be interrelated or even conflicting. To address the ethical considerations and issues, designers need to discover ethical values from stakeholders and related communities to understand their impacts on the individuals and society (Flanagan et al., 2008). By answering this research question, we aim to identify a list of ethical values, examine their relationships, and explore methods for eliciting them from stakeholders.
- **RQ3:** How can stakeholders' requirements and their values be addressed in the design and development of software systems? And how can ethical values be translated into software requirements? **Rationale:** To ensure that ethical values are reflected in the design and development of software systems, it is essential to translate these values into concrete terms and actionable requirements. By answering this research question, we aim to recognize the approaches,[5] methods,[6] and techniques[7] that support and embed the stakeholders' values in software requirements, design and development. This process involves aligning ethical considerations with technical specifications to ensure that software systems reflect the identified ethical values.

3.2. Search strategy

One of the main motivations of this SLR is to identify primary studies addressing the specified research questions by using a well-defined search strategy. We present the selected source and search strings for our research in the following.

Source. The choice of Google Scholar as our meta search engine was due to Jacsó (2005): (i) the reputation of being a reliable scholarly database and (ii) the power to obtain a broader range of sources (*i.e.,* articles) by encompassing other libraries, such as IEEE Xplore, and Scopus. Research has also shown that Google Scholar has the highest intersection when compared to Web of Science and Scopus (Martín-Martín et al., 2018b).

Search string. We formulated our meta search strings based on the identified keywords and the main intent of the research questions (*i.e.,* focusing on key stakeholders, ethical values, and value sensitive design); we also took synonyms of these terms into account. To ensure comprehensive results, we included both "ethics" and "ethical" in our search strings, while "moral" and "morality" were also embedded in

[5] Defined as the set of assumptions, theories, and working ideas held by individuals concerned with their field (Burnham, 1992; Hofler, 1983).

[6] Defined as patterns and ways of working based on the approach (Burnham, 1992; Hofler, 1983).

[7] Defined as practice activities, tools, procedures, or strategies that are used to implement the method (Burnham, 1992; Hofler, 1983).

the query. Ethics and morality are two sides of the same coin, and many studies used them as synonyms. In contrast, ethics and morals/morality are different in nature (morality refers to generic principles set by a group, while ethics refers to a response to a specific situation) (Williams and Lear, 2011). Furthermore, given the multidisciplinary nature of ethics, it is essential to specify the domain under consideration. Regarding the objective of this SLR which is analyzing ethical aspects and issues in software-intensive systems, our main focus is on the generic level related to these systems (*i.e.,* software architecture and software design in general, rather than any specific phase in particular, like requirement engineering or testing). Therefore, the search string should include phrases such as "software engineering", "software architecture", "software development", "software design", or "software system" to discover more relevant literature. To select a representative yet feasible search query, we proceeded as follows. First, we conducted a pilot search with the different combinations of defined terms in a specified time period (from 1980 to 2022). Some of them are elaborated on below.

Query 1 `(''ethics'' OR ''ethical'' OR ''moral'' OR ''morality '') AND ''software'' AND (''engineering'' OR ''architecture'' OR ''development'' OR ''design'' OR ''system'')`

This set of terms was discarded as it resulted in too many hits (approximately 3.3 million). To support search feasibility, we decided to limit the search to the title of the studies. So, the reasonable combination with these terms was the following form with 229 hits.

Query 2 `(intitle:''ethics'' OR intitle:''ethical'' OR intitle:''moral'' OR intitle:''morality'') AND intitle:''software'' AND (intitle: ''engineering'' OR intitle:''architecture'' OR intitle:''development'' OR intitle:''design'' OR intitle: ''system'')`

Also, as "value" is a motivation or foundation for one's morality, we need to apply it in our research query. However, combining this term even with Query 2 can result in a large number of irrelevant studies that chiefly refer to the economic worth. Therefore, based on the definition of value used in this study (*i.e.,* as something that is important to a group or an individual), we defined a new query including phrases like "value-sensitive" and "value-centered". This query led to 430 hits. Based on the similar reason in creating Query 2, *i.e.,* avoiding producing an enormous number of irrelevant studies, these phrases were only applied in the title of the studies as below.

Query 3 `intitle:value AND (intitle:sensitive OR intitle:centered) AND (intitle:design OR intitle: software)`

In total, we considered both Queries 2 and 3 as the search strings for our research, resulting in a total of 623 hits after removing duplications.

3.3. Selection criteria

Following the guidelines outlined in Keele et al. (2007) to conduct an SLR in software engineering, we established explicit inclusion and exclusion criteria to guide an objective selection of the relevant primary studies. Inclusion was based on studies meeting all the inclusion criteria, while exclusion involved studies that met at least one of the exclusion criteria.

Inclusion criteria. The established inclusion criteria were:

IC1: Studies that were peer-reviewed. With this inclusion criterion, we ensure that only papers with enough level of quality are included.

IC2: Studies were available through digital libraries.

IC3: Studies written in English.

IC4: Studies that focused on ethics and values in software engineering, software architecture, software design, software development, and software systems. This inclusion criterion is applied to select studies whose main concerns are ethical aspects in the design and development process of software systems or technologies.

Exclusion criteria. The considered exclusion criteria were:

EC1: Studies that were not subject to peer review and not original research or/and unpublished. Considering this criterion, discard studies that lack a high level of quality.

EC2: Studies in the form of books/book sections, editorials, tutorials, short papers, reports, thesis, posters, systematic reviews, and tertiary studies. This exclusion criterion is adopted to exclude studies that do not provide the desired level of information in terms of details and elaborations about ethics in software engineering.

EC3: Studies that we could not find a full version of them.

EC4: Studies that were not written in English, as they could be challenging to analyze and also took considerable time to translate.

EC5: Studies that did not focus on ethics or ethical aspects (*i.e.,* they just concentrated on the context of software design or development process).

EC6: Studies that did not focus on software design or development process (*i.e.,* the studies whose primary attention was on ethics or ethical aspects). For example, studies focused on the ethical principles or code of ethics in software engineering. Also, studies were about ethics in the curricula context.

EC7: Studies that were generic and did not have case studies or information related to our research questions.

3.4. Data extraction and synthesis

In this section, we provide an explanation of the codes used for the data synthesis and the coding procedure employed for analyzing and coding the studies.

Coding for data synthesis. To address our research questions, (*i.e., RQ*1, *RQ*2, and *RQ*3), we identify (see Table 1) the specific data items that we need to extract from the primary studies.

The code pertaining to the year of publication, referred to as the "publication year", and the code associated with the venue, known as the "venue", aim to provide insights into how primary studies in SE ethics are distributed among various years and publication venues.

The code related to stakeholders, *i.e.,* "stakeholder types" and "stakeholder concerns" are intended to answer the *RQ*1. More precisely, the collected data from these codes can be used by software engineers to answer questions in this regard, such as (i) who are the individuals that affect or be affected by software-intensive systems (both positively or negatively)?, (ii) what are their concerns from an ethical point of view? And so on.

Furthermore, by defining codes such as "values" and "value relations", and "value elicitation" we aim to answer the *RQ*2 that focuses on the values in this context. Extracting data through these codes and analyzing them can help software engineers to gain insights into (ethical) values and enable them to answer some questions in this regard, such as (i) what are the important ethical values that need to focus on during the software design and development process?, (ii) what are the relations among the values?, (iii) what methods can be utilized to elicit values from stakeholders? And so on.

Lastly, we define the codes of "embedding values of stakeholders in the software design process" and "mapping values to requirements" to identify how requirements and values of different stakeholders can be addressed in software design and development. We need to know what are approaches, methods, techniques, *etc.*, which support and embed the stakeholders' values in the software design process. Also, we need to gain insight into the goals and the activities that these approaches, methods, techniques, *etc.*, conduct or do neglect. The collected data through these codes help software engineers answer the *RQ*3 and also some related questions in this regard, for example: (i) what are the goals of these approaches, methods, techniques, *etc.*?, (ii) what are their important activities?, (iii) how can values be converted into software requirements? And so on.

Table 1
Data items extracted from each selected study.

Data item	Description	Relevant RQ
Publication year	The year of publication for the study.	Study overview
Venue	The venue of publication for the study, which can be a journal, conference, or workshop.	Study overview
Stakeholder types, including: • Those who use the system with direct benefit/harm. • Those who interact with the system with indirect benefit/harm. • Those who are not involved in the system's development or use and may have or not have benefit/harm from the system.	Stakeholders are people who could either affect or be affected by the software systems, for example, end users of the system, designers, and society at large. These individuals can be further classified based on their relationship with the system into two main categories: (i) people who have a direct relation with the system, use or interact with the system, and (ii) people who do not use or interact with the system, but they are affected by the system or its output through others.	RQ1
Stakeholder concerns	The issues (*i.e.,* harms, concerns, challenges, problems, risks, *etc.*.) that stakeholders care about and the issues that can be tackled by analyzing and embedding ethical values throughout the software design and development process, for example, privacy violation.	RQ1
Ethical values	Values are expressions of what humans, organizations, *etc.*, find important and some conception of what they consider good, bad, right, and wrong. Values can be context-dependent or can be instantiated in specific situations, but often are formulated abstractly, such as privacy and justice.	RQ2
Value elicitation	There are methods and techniques to extract values from the stakeholders, *e.g.,* interviews and surveys.	RQ2
Value relations, including: • Conflict among values • Tension among values • Congruity among values	The way values interrelate with each other, for example, conflict, tension, and congruence.	RQ2
Embedding values of stakeholders in the software design process	Recognizing the approaches, methods, techniques, *etc.*, which support and embed the stakeholders' values in the software design process.	RQ3
Mapping values to requirements	The approaches to translate values into design requirements, for example, value hierarchy.	RQ3

Coding procedure. After selecting the list of primary studies, we proceeded to synthesize and analyze their data using the coding method proposed by Miles and Huberman (1994). This procedure was started by analyzing 20 selected primary studies to answer each research question. Accordingly, we formed a data extraction sheet that contains data items like stakeholder types, values, and the like, as previously mentioned. To verify the reliability of the extraction process, we piloted the data extraction sheet with this initial set of studies. This piloting involved testing the extraction process on a smaller, representative sample to identify any issues or inconsistencies in the coding and extraction methods. By refining our approach during the pilot, we ensured that our process was robust before applying it to all the primary studies. Additionally, we aimed for a homogeneous set of studies and reduced bias by extracting more qualitative details. The extracted data items evolved as the research progressed and our understanding of the area deepened. VSD methods provided the theoretical underpinning for our review's codification, guiding our focus and coding efforts on stakeholders, values, and other relevant aspects, as detailed below.

This is the manner in which the studies were analyzed and coded:

Step 1: We performed coding to ascertain the years of publication and the types of venues mentioned in the studies.

Step 2: Each study was coded based on the involved stakeholders related to software-intensive systems in various domains. This coding process utilized three roles of stakeholders: (i) stakeholders that use the system and receive direct benefit/harm from the system, (ii) stakeholders that have indirect benefits/harms in interaction with the system, and (iii) stakeholders that do not use or interact with the system, they may or may not receive benefit/harm from the system. If the selected studies explicitly mentioned stakeholders, they were immediately classified into these categories. In cases where stakeholders were not explicitly mentioned, we interpreted and classified them based on their relations with the system (*i.e.,* using or interacting with the system). Second, the concerns of stakeholders who were affected by or affected the system, as discussed in the studies, were then coded. To accomplish this, the coding procedure was conducted iteratively, with the identification of indicators such as issues, harms, concerns, challenges, problems, risks, *etc.*, to extract and code the stakeholders' concerns.

Step 3: We conducted coding to identify the values mentioned in the studies. For each value elicited from the studies, we listed its definition(s) along with the relations between them. Additionally, we coded any methods and techniques used to extract those values from the studies.

To classify and code the values, we used the human value structure of Schwartz as the underpinning theoretical framework (Schwartz, 2012, 1992). We initiated our categorization process by employing the generic value categories and sub-values of this structure as the start list. In our analysis, we first extracted the values from the studies and analyzed their meaning. Recognizing that one value can be described using different terminologies, we opted to leverage a meta-inventory of human values (in Cheng and Fleischmann (2010)), to reduce ambiguity within the value categories. This meta-inventory integrates the categorized values into a cohesive framework, facilitating a more comprehensive exploration of human values. Utilizing this framework, we assigned labels that encapsulated similar meanings. For instance, we used the label "responsibility" to encompass values such as "accountability" and "responsibility". With the assistance of this framework, we proceeded to map the values according to Schwartz's list of values. Throughout the iterations, some extracted values were directly mapped to the values in this list (shown with red in Fig. 2), while some values could not be directly mapped (shown with blue in Fig. 2). Also, some values in this structure were not mentioned in the primary studies (shown with black in Fig. 2).

Step 4: We coded the approaches, methods, techniques, *etc.*, reported in the studies, which used to address ethical considerations and to embed the stakeholders' requirements and values during the software design and development process. As all the primary studies utilized the VSD approach to account for ethical values, we focused on this

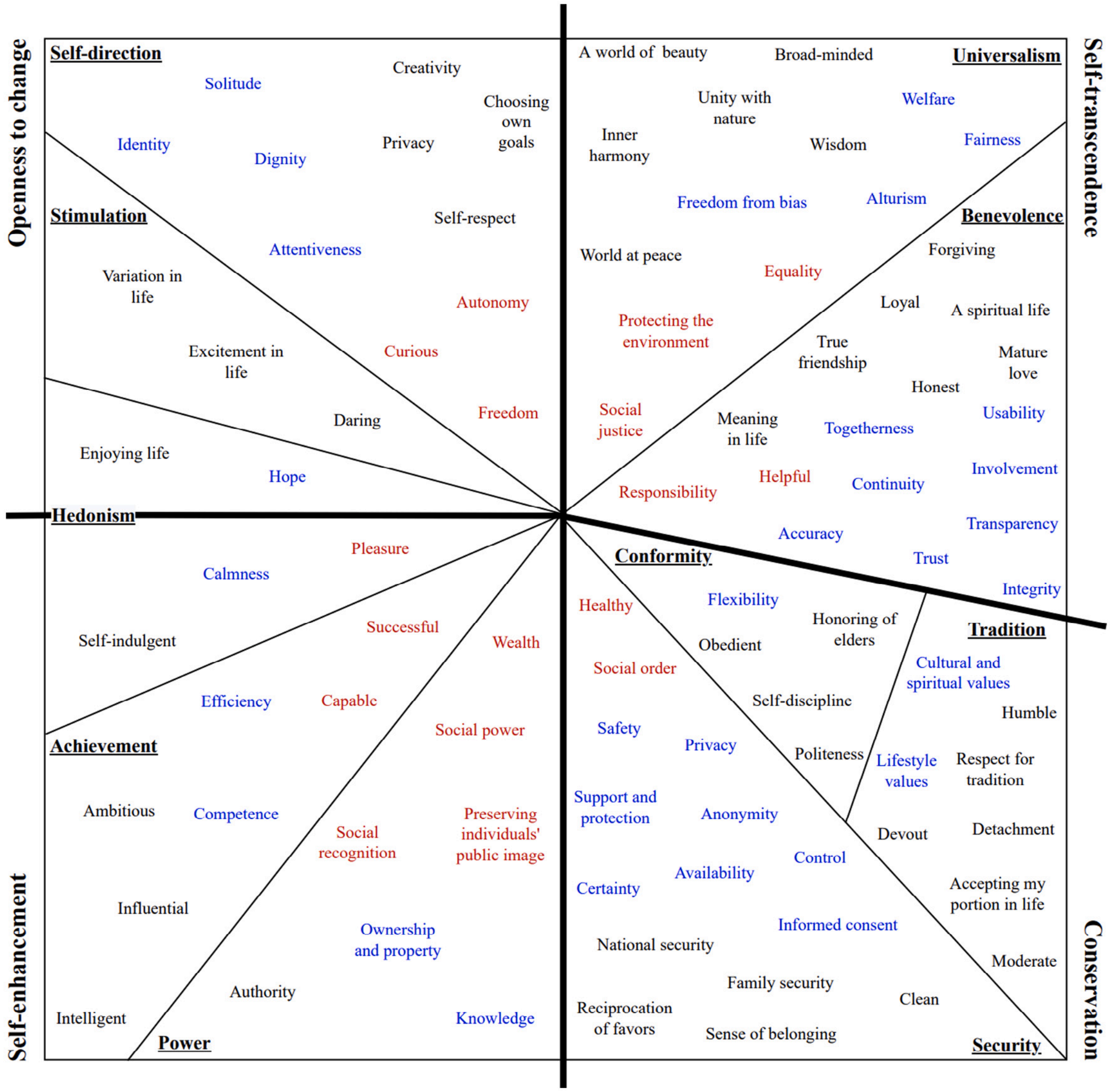

Fig. 2. Categorizing the extracted values from the primary studies based on Schwartz value structure (Schwartz, 2007, 1992). The values shown in **red** indicate the extracted values from the primary studies that already exist in Schwartz's structure, and we found examples in our SLR. The values in **blue** are the values that we added from our SLR to the values of Schwartz's structure, and the values in **black** are values that were in this structure and we did not find any in our primary studies. Thick lines shown in the structure specify the orthogonal dimensions: (i) self-enhancement (including hedonism, achievement, and power) vs. self-transcendence (including universalism and benevolence) and (ii) openness to change (including self-direction, stimulation, and hedonism) vs. conservation (including tradition, conformity, and security). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

approach and its activities. According to the three investigations in this approach (as shown in Fig. 1), we coded the existing activities in each investigation, such as "potential value tensions and conflicts" and "translation of values into design requirements". Studies explicitly mentioning these activities were categorized into activities "$C1$-$C5$", "$E1$-$E3$", or "$T1$-$T3$"; otherwise, we interpreted and classified them accordingly.

3.5. Conducting the systematic literature review

Fig. 3 provides an overview of the research process we detailed in the previous subsections. The following provides the key information related to executing it.

We conducted our search using Google Scholar as the database in September 2022. The two search strings defined in Section 3.2, were executed on this database by adapting to the search engine settings and constraints. The first search string (focusing on ethics and morality in software design and development) led to 229 studies. The second search string (focusing on value in software design and development) led to 430 studies. After eliminating duplicate studies, we identified a total of 623 studies. After applying the selection criteria, we included 85 primary studies in our final data set. Then, we employed coding techniques to extract information from each included study.

The results we report are about publication trends, stakeholder identification and concerns, value identification and value relations, as well as the incorporation of stakeholders' requirements and values in software design and development. We discussed potential threats to the study results.

4. Results

The search and selection process resulted in 85 primary studies that we analyzed as described in Section 3. In the following, we present a general overview of the publication trends and the results addressing our research questions.

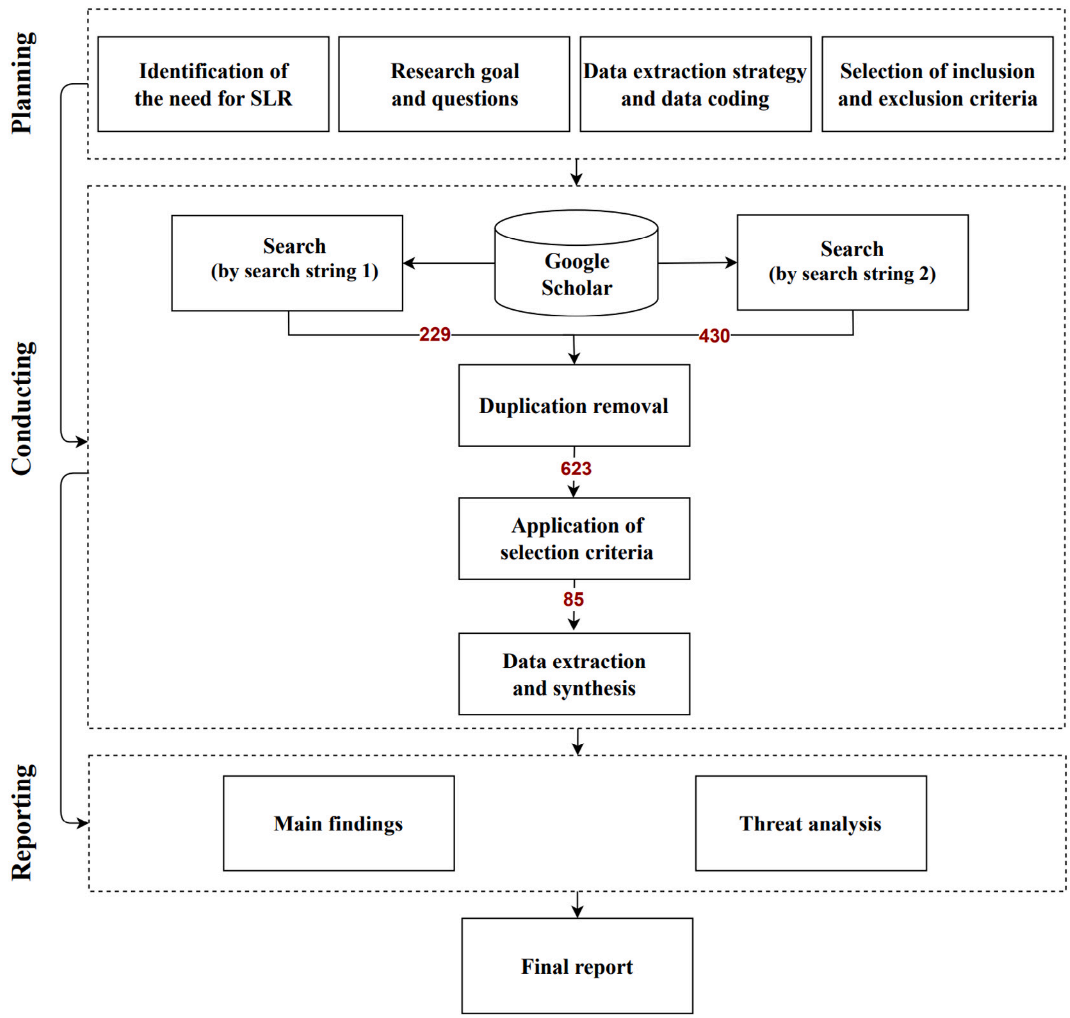

Fig. 3. Overview of the research process.

4.1. Publication trends

In this section, we aim to discover the published efforts and trends in SE ethics based on publication year, publication venue, and venue type.

Over the last decades, there has been a modest yet visible growth of publications in the context of SE ethics. Fig. 4 shows the relationship between publication year and venue type of the selected studies, and the number in bubbles represents the corresponding number of studies. As shown, the primary studies span from 2000 to 2022. Also, we observe a modest yet visible growth in recent years.

Journal and conference are the most targeted venue types by the studies. Journal and conference are the two main venue types for the selected studies, which account for 49.4% (42 out of 85 studies) and 44.7% (38 out of 85 studies), respectively.

Some publication venues publish more articles on SE ethics. Ethics has a multidisciplinary nature, which has attracted attention from various fields like social sciences and software engineering. Table 2 in online Appendix B[8] gives an overview of the studies in SE ethics with respect to publication venues and venue types. Based on the findings, the 85 primary studies were distributed over 64 different publication venues. Upon analysis, it becomes apparent that there is significant activity in various publication venues regarding software engineering ethics. Among these venues, the majority (53 venues, 62.3%) have published only one paper each. Exceptions are the journal "Science and Engineering Ethics", which published seven papers, and "Conference on Human Factors in Computing Systems (CHI)" published four papers on software engineering ethics.

[8] To further explore publication venues, the interested reader can find Table 3 in the replication package available online at https://github.com/RaziehAlidoosti/Exploring-Ethical-Values-in-Software-Systems-a-Systematic-Literature-Review.git.

4.2. RQ1: Identifying stakeholders

One of the important objectives in designing software systems is satisfying stakeholders through supporting their values and meeting their needs. The first step in achieving this goal is to identify the stakeholders who affect or are affected by a system and identify their concerns.

4.2.1. Stakeholder roles

In software engineering, stakeholders are the people who use the system, build the system, or are affected by the system. From the primary studies, we identified individuals, groups, and institutions reported to use, build, and be affected by the software system. We further categorized those stakeholders based on the way they interact with the system (*i.e.,* use or build) and are affected by the system. Our results suggest three overarching roles of stakeholders, shown in Fig. 5: (i) *system users*—those who directly use the system, (ii) *system development organisation*—those that design and develop the system, and (iii) *indirect*

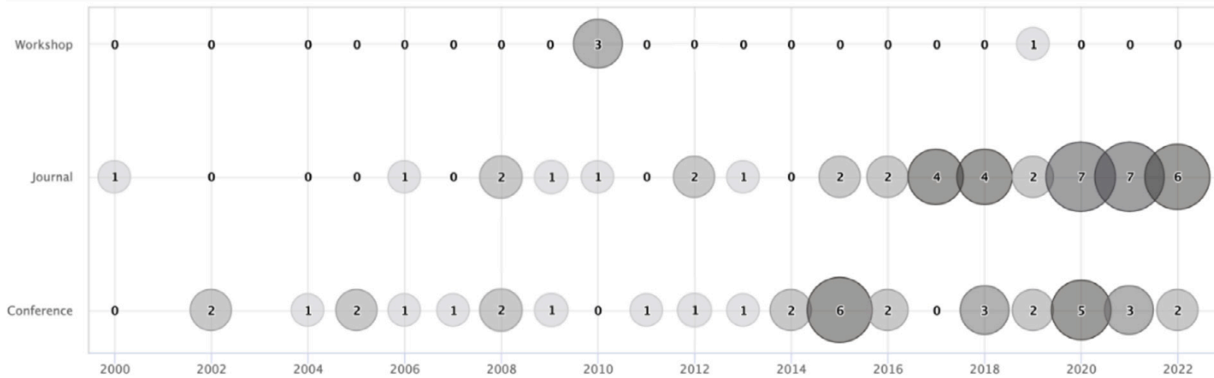

Fig. 4. Number of primary studies based on publication year and venue type.

stakeholders—those who are indirectly affected by the system. Further, these three stakeholder roles can be specialized in more specific roles, which we call *role types*. For example, engineer is a role type that falls under the role of system development organization. The roles, along with their types discovered in the studies, are summarized in Table 2.

System users are individuals, groups, or institutions that use the system and receive direct benefit/harm from the system. Consider Electronic Health Record (EHR) system for use in hospitals. Hospital personnel such as doctors, in-take receptionists as well as insurance companies directly use the system; hence, they are 'system users'. There was distinguished different kinds of system users in the primary studies. The most prevalent one, is *end users* (mentioned in 73 out of 85 studies). Some primary studies also recognize institutions as system users, *e.g., organizations* or *government*.

System development organisation are those individuals, groups, or institutions that make decisions, design, develop, fund, and support the system, and although they may not use the system, they affect and are affected by it. When designing and developing the software, this type of stakeholders, explicitly or implicitly, impose their own individual values and the organizational values in the system. They may also be affected by the system, for instance, through the success or failure of the project. Continuing with the example of the EHR system, the personal and organizational values of product owners, software designers, project managers, and the software organization as a whole affect the design of EHR. Invariably, the success or failure of the EHR system affects these stakeholders as well.

System development organization, in the roles of *designers* and *developers*, were identified as the most prevalent ones (mentioned in 13/85 and 13/85 studies, respectively).

Indirect stakeholders are the individuals, groups, or institutions that are indirectly affected by a software system but never interact with the system. In the example of the EHR system, indirect stakeholders may be the patients and their families who are dependent upon information in the software system but do not use it directly. The most prevalent indirect stakeholders identified in the primary studies are *society* and *third parties* (mentioned in 18/85 and 10/85 studies, respectively).

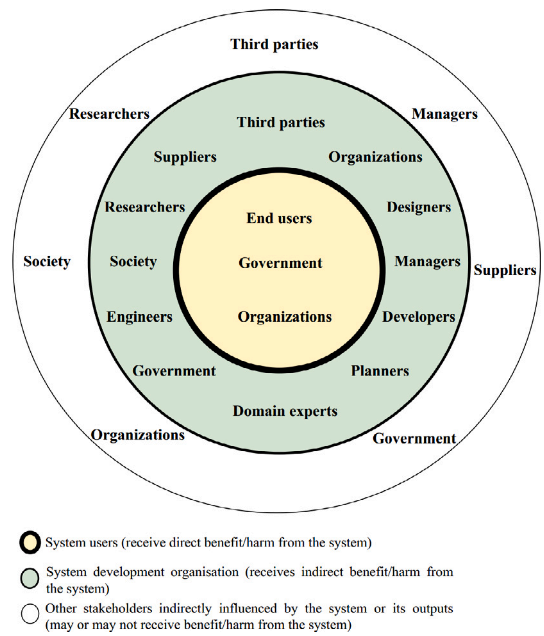

Fig. 5. Stakeholder roles and their role types in terms of receiving benefit/harm from the system.

4.2.2. Stakeholders' concerns

The concerns of stakeholders are an expression of specific interest about some topics related to a particular context that stakeholders care about Hilliard (1999). To incorporate the stakeholders' values in software design, their concerns should be understood and considered (Vermaas et al., 2015). Some of their concerns may have importance from the ethical point of view that could affect ethical values. For example, stakeholders may have concerns about privacy violations that could affect the ethical values of privacy and security. So, there is a need to address the stakeholders' concerns to support and embed the ethical values and their corresponding software requirements in the design and development of software systems. Regarding the three mentioned stakeholder roles, each has different ethical concerns with respect to their relationship with the system, and thus, different values may need to be embedded in the software design. Based on the primary studies, however, we found concerns of two stakeholder roles surrounding software systems, *i.e.,* system users and system development organization. Table 7 (in online Appendix D[9]) shows these concerns along with their relevant values.

Stakeholders are mainly concerned with the issues related to privacy, safety, and trust in software systems. All stakeholders (who affect or are affected by the systems) have different kinds of concerns about software systems. These concerns are about the values at stake or the harms of the system towards stakeholders. By drawing on these concerns, we found that they are most frequently related to the values of *privacy*, *safety*, and *trust*, mentioned in 16, 11, and 11 out of 85 studies, respectively.

The two values of privacy and safety fall under the value category of security, representing 30 studies (35.3%), while trust belongs to the category of benevolence, representing 20 studies (23.5%). We have also found other stakeholders' concerns in the primary studies relevant

[9] For the interested reader, Table 7 is available here.

Table 2
Three stakeholder roles and their role types.

Stakeholder role	Role definition	Role type	Description	Studies
System users	Stakeholders that use the system and receive direct benefit/harm from the system.	End users	Individuals who systems are designed for them and ultimately use systems.	Alsheikh et al. (2011), Miller et al. (2007), Betz and Fritsch (2016), Alvarenga and Tanev (2017), Walldius et al. (2005), Yoo et al. (2013), Nouwen et al. (2015), Zaman and JafariNaimi (2015), Xu et al. (2012), Detweiler et al. (2010b), Friedman and Kahn (2000), Umbrello and De Bellis (2018), Le Dantec et al. (2009), Xu et al. (2008), Briggs and Thomas (2015), Sambasivan and Jackson (2008), Simons and Verhagen (2008), Umbrello (2019), Badillo-Urquiola et al. (2020), Denning et al. (2014), Borning et al. (2004), Royakkers and Steen (2017), Friedman et al. (2006), Dadgar and Joshi (2015a), Maathuis et al. (2019), Barn and Barn (2018), Dadgar and Joshi (2015b), Friedman et al. (2002), Borning et al. (2005), Sackey (2020), Detweiler et al. (2010a), Nouwen and Zaman (2018), Walldius et al. (2015), Mithun et al. (2018), Le Dantec and Poole (2008), Ghosh (2016), Grünloh (2018), Watkins (2018), Burrows et al. (2015), Friedman and Borning (2002), Thornton et al. (2018), Rychwalska and Roszczynska-Kurasinska (2017), Heger (2019), Cawthorne and Cenci (2019), Harbers and Neerincx (2017, 2014), Harbers et al. (2010), Walton and DeRenzi (2009), Detweiler and Hindriks (2012), Van Andel et al. (2015), Lüthi et al. (2021), Umbrello and Yampolskiy (2022), Kumari et al. (2021), Helbing et al. (2021), Primiero et al. (2020), Graubohm et al. (2020), Iversen et al. (2020), Cruz-Martínez et al. (2021), Tokranova (2020), Köhler et al. (2022), Helbing et al. (2021), Sapraz and Han (2021), García and Gil (2020), Smits et al. (2022), Kowe et al. (2022), Umbrello et al. (2021), Oktay and Pender (2020), Dexe et al. (2020), Van Wynsberghe (2012), Jaljolie et al. (2022), Umbrello and Van de Poel (2021), Cenci and Cawthorne (2020), Umbrello (2020)
		Organizations	A group of individuals who work together to achieve a common goal.	Vermaas et al. (2010), Royakkers and Steen (2017), Maathuis et al. (2019), Cawthorne and Cenci (2019), Walton and DeRenzi (2009), Jaljolie et al. (2022)
		Government	A group of individuals that govern society through legislation and regulatory rules.	Maathuis et al. (2019), Sapraz and Han (2021), Jaljolie et al. (2022)
System development organization	Stakeholders that are involved in software design and development, they receive indirect benefit/harm from the system.	Designers	Individuals who use the designing techniques and methods to create a foundation for developing software systems.	Nouwen et al. (2015), Zaman and JafariNaimi (2015), Umbrello and De Bellis (2018), Xu et al. (2008), Umbrello (2019), Badillo-Urquiola et al. (2020), Borning et al. (2004), Vermaas et al. (2010), Cheon and Su (2016), Sackey (2020), Heger (2019), Harbers et al. (2010), Umbrello and Yampolskiy (2022)
		Developers	Individuals like programmers and coders who are involved in the design and development process of software systems.	Badillo-Urquiola et al. (2020), Vermaas et al. (2010), Friedman et al. (2006), Dadgar and Joshi (2015a), Friedman et al. (2002), Sackey (2020), Friedman and Borning (2002), Heger (2019), Van Andel et al. (2015), Lüthi et al. (2021), Graubohm et al. (2020), Burmeister et al. (2021), García and Gil (2020)
		Domain experts	Individuals who have knowledge and expertise in a specific domain like software development.	Nouwen et al. (2015), Zaman and JafariNaimi (2015), van de Kaa et al. (2020), Harbers and Neerincx (2017), Vernim et al. (2022), Boyd (2022), Cruz-Martínez et al. (2021), Tokranova (2020), Köhler et al. (2022), Kowe et al. (2022)
		Organizations	A group of individuals who work together to achieve a common goal.	Nouwen et al. (2015), Zaman and JafariNaimi (2015), Xu et al. (2008), Nouwen and Zaman (2018), Vernim et al. (2022), Kumari et al. (2021), García and Gil (2020), Jaljolie et al. (2022)
		Suppliers	Individuals who provide necessary resources for organizations to meet a goal or to take action *e.g.,* hardware and software for software development teams to build software systems.	Alvarenga and Tanev (2017), Denning et al. (2014), Cawthorne and Cenci (2019), Vernim et al. (2022), Umbrello and Yampolskiy (2022), García and Gil (2020)
		Managers	Individuals who coordinate human resources, activities, and processes in organizations to *e.g.,* develop software systems.	Nouwen et al. (2015), Zaman and JafariNaimi (2015), Heger (2019), Vernim et al. (2022), Boyd (2022)
		Engineers	Individuals who analyze, design, and build systems and structures to accomplish objectives with considerations of limitations, *e.g.,* software engineers who are responsible for applying software engineering principles in the design and development of software systems.	Watkins (2018), Heger (2019), Umbrello and Yampolskiy (2022), Boyd (2022)
		Researchers	Individuals who collect, analyze, and interpret data to explore and solve issues in a specific context like software systems.	Denning et al. (2014), Van Andel et al. (2015), Helbing et al. (2021)
		Government	A group of individuals that govern society through legislation and regulatory rules.	Harbers et al. (2010), Helbing et al. (2021), Jaljolie et al. (2022)
		Planners	Individuals who create plans and evaluate individuals' requirements (like systems' stakeholders) to ensure meet their needs.	Watkins (2018), Heger (2019)
Other stakeholders indirectly influenced by the system or its outputs	Stakeholders that do not interact with the system (*i.e.,* use or build), they may or may not receive benefit/harm from the system.	Society	A group of individuals who live together and have continuous social interaction.	Borning et al. (2004), Royakkers and Steen (2017), Cawthorne and Robbins-van Wynsberghe (2019), Detweiler et al. (2010a), Mithun et al. (2018), Thornton et al. (2018), Cawthorne and Cenci (2019), Harbers and Neerincx (2014), Walton and DeRenzi (2009), Boyd (2022), Helbing et al. (2021), Graubohm et al. (2020), Iversen et al. (2020), Köhler et al. (2022), García and Gil (2020), Dexe et al. (2020), Jaljolie et al. (2022)
		Third parties	Organizations or individuals that play an important role in projects (like software projects) but are not the ones that directly involved in.	Miller et al. (2007), Alvarenga and Tanev (2017), Denning et al. (2014), Friedman et al. (2006), Walton and DeRenzi (2009), Van Andel et al. (2015), Graubohm et al. (2020), García and Gil (2020), Dexe et al. (2020), Jaljolie et al. (2022)
		Organizations	A group of individuals who work together to achieve a common goal.	Alvarenga and Tanev (2017), Simons and Verhagen (2008), Borning et al. (2004), Van Andel et al. (2015), Lüthi et al. (2021), Vernim et al. (2022), Graubohm et al. (2020), Dexe et al. (2020)
		Suppliers	Individuals who provide necessary resources for organizations to meet a goal or to take action *e.g.,* hardware and software for software development teams to build software systems.	Alvarenga and Tanev (2017), Simons and Verhagen (2008), Denning et al. (2014), Mithun et al. (2018), Detweiler and Hindriks (2012), Graubohm et al. (2020), Dexe et al. (2020)
		Government	A group of individuals that govern society through legislation and regulatory rules.	Simons and Verhagen (2008), Detweiler et al. (2010a), Cawthorne and Cenci (2019), Graubohm et al. (2020), Helbing et al. (2021), Dexe et al. (2020)
		Managers	Individuals who coordinate human resources, activities, and processes in organizations to *e.g.,* develop software systems.	Miller et al. (2007), Betz and Fritsch (2016), Lüthi et al. (2021), Vernim et al. (2022), Dexe et al. (2020), Jaljolie et al. (2022)
		Researchers	Individuals who collect, analyze, and interpret data to explore and solve issues in a specific context like software systems.	Denning et al. (2014), Detweiler et al. (2010a), Helbing et al. (2021), Jaljolie et al. (2022)

Table 3
Extracted values from the primary studies based on the Schwartz value structure. The values shown in "red" indicate the extracted values from the primary studies that already exist in Schwartz's structure, and the values in "blue" indicate the values that we added from our SLR to the values of Schwartz's structure.

Schwartz's dimension	Schwartz's value category	Extracted values from the studies	Studies
Conservation	Security (SE)	Healthy	Friedman and Kahn (2000), Friedman et al. (2006), Cawthorne and Robbins-van Wynsberghe (2019), Cummings (2006), Cawthorne and Cenci (2019), Van Andel et al. (2015)
		Social order	Rychwalska and Roszczynska-Kurasinska (2017)
		Privacy	Alsheikh et al. (2011), Miller et al. (2007), Betz and Fritsch (2016), Alvarenga and Tanev (2017), Xu et al. (2012), Detweiler et al. (2010b), Friedman and Kahn (2000), Le Dantec et al. (2009), Xu et al. (2008), Simons and Verhagen (2008), Umbrello (2019), Badillo-Urquiola et al. (2020), Denning et al. (2014), Friedman et al. (2006), Dadgar and Joshi (2015a), Maathuis et al. (2019), Cawthorne and Robbins-van Wynsberghe (2019), van de Kaa et al. (2020), Barn and Barn (2018), Friedman et al. (2002), Detweiler et al. (2010a), Grünloh (2018), Watkins (2018), Burrows et al. (2015), Harbers and Neerincx (2017), Detweiler and Hindriks (2012), Gan and Moussawi (2022), Lüthi et al. (2021), Oktay and Pender (2020), Jaljolie et al. (2022), Primiero et al. (2020), Graubohm et al. (2020), Iversen et al. (2020), Smits et al. (2022), Burmeister et al. (2021), García and Gil (2020), Köhler et al. (2022), Umbrello (2020), Tokranova (2020), Kowe et al. (2022)
		Safety	Miller et al. (2007), Alvarenga and Tanev (2017), Yoo et al. (2013), Nouwen et al. (2015), Zaman and JafariNaimi (2015), Umbrello and De Bellis (2018), Xu et al. (2008), Briggs and Thomas (2015), Badillo-Urquiola et al. (2020), Denning et al. (2014), Royakkers and Steen (2017), Cawthorne and Robbins-van Wynsberghe (2019), van de Kaa et al. (2020), Cummings (2006), Cheon and Su (2016), Nouwen and Zaman (2018), Ghosh (2016), Watkins (2018), Kumari et al. (2021), Wambsganss et al. (2021), Graubohm et al. (2020), Iversen et al. (2020), Smits et al. (2022), Sapraz and Han (2021), Köhler et al. (2022), Kowe et al. (2022)
		Availability	Xu et al. (2012), Briggs and Thomas (2015), Umbrello (2019), Denning et al. (2014), Friedman et al. (2006), Watkins (2018), Friedman and Borning (2002), Harbers and Neerincx (2014), Walton and DeRenzi (2009), Van Andel et al. (2015), Lüthi et al. (2021), Jaljolie et al. (2022), Graubohm et al. (2020), Dexe et al. (2020), Smits et al. (2022), Umbrello (2020), Cruz-Martínez et al. (2021)
		Informed consent	Miller et al. (2007), Friedman and Kahn (2000), Le Dantec et al. (2009), Umbrello (2019), Friedman et al. (2006, 2002), Cummings (2006), Friedman and Borning (2002), Jaljolie et al. (2022), García and Gil (2020), Sapraz and Han (2021), Tokranova (2020), Kowe et al. (2022)
		Control	Nouwen et al. (2015), Xu et al. (2012), Umbrello (2019), Maathuis et al. (2019), Sackey (2020), Nouwen and Zaman (2018), Harbers and Neerincx (2014), Kowe et al. (2022)
		Support and protection	Alsheikh et al. (2011), Badillo-Urquiola et al. (2020), Harbers and Neerincx (2014), Lüthi et al. (2021), Kumari et al. (2021), Köhler et al. (2022), Cruz-Martínez et al. (2021)
		Anonymity	Miller et al. (2007), Detweiler et al. (2010b), Le Dantec et al. (2009), Detweiler et al. (2010a), Barn and Barn (2018), Heger (2019)
		Certainty	Heger (2019), Umbrello and Yampolskiy (2022), Jaljolie et al. (2022)
	Conformity (CO)	Flexibility	van de Kaa et al. (2020), Lüthi et al. (2021), Boyd (2022), Graubohm et al. (2020), Köhler et al. (2022), Kowe et al. (2022)
	Tradition (TR)	Cultural and spiritual values	Alsheikh et al. (2011), Cheon and Su (2016)
		Lifestyle values	Burrows et al. (2015), Cruz-Martínez et al. (2021)
Openness to change	Self-direction (SD)	Autonomy	Nouwen et al. (2015), Le Dantec et al. (2009), Sambasivan and Jackson (2008), Badillo-Urquiola et al. (2020), Dadgar and Joshi (2015a), Maathuis et al. (2019), van de Kaa et al. (2020), Barn and Barn (2018), Dadgar and Joshi (2015b), Friedman et al. (2002), Nouwen and Zaman (2018), Ghosh (2016), Grünloh (2018), Watkins (2018), Thornton et al. (2018), Rychwalska and Roszczynska-Kurasinska (2017), Heger (2019), Cawthorne and Cenci (2019), Walton and DeRenzi (2009), Detweiler and Hindriks (2012), Van Andel et al. (2015), Gan and Moussawi (2022), Lüthi et al. (2021), Oktay and Pender (2020), Primiero et al. (2020), Umbrello et al. (2021), Umbrello and Van de Poel (2021), Smits et al. (2022), Burmeister et al. (2021), García and Gil (2020), Sapraz and Han (2021), Köhler et al. (2022), Tokranova (2020), Kowe et al. (2022), Cruz-Martínez et al. (2021)
		Freedom	Le Dantec and Poole (2008), Rychwalska and Roszczynska-Kurasinska (2017), Gan and Moussawi (2022), Jaljolie et al. (2022), Graubohm et al. (2020), Kowe et al. (2022)
		Curious	Rychwalska and Roszczynska-Kurasinska (2017), Harbers and Neerincx (2014)
		Dignity	Nouwen et al. (2015), Le Dantec et al. (2009), Cheon and Su (2016), Nouwen and Zaman (2018), Heger (2019), Detweiler and Hindriks (2012), Vernim et al. (2022), Gan and Moussawi (2022), Oktay and Pender (2020), Jaljolie et al. (2022), Cenci and Cawthorne (2020), Kowe et al. (2022)
		Identity	Le Dantec et al. (2009), Le Dantec and Poole (2008), Cawthorne and Cenci (2019), Harbers and Neerincx (2014), Vernim et al. (2022), Lüthi et al. (2021), Smits et al. (2022), Köhler et al. (2022), Tokranova (2020)
		Attentiveness	Van Wynsberghe (2013), Friedman et al. (2006), Van Wynsberghe (2012)
		Solitude	Friedman et al. (2006)
	Hedonism (HE)	Pleasure	Helbing et al. (2021), Kowe et al. (2022)
		Calmness	van de Kaa et al. (2020), Watkins (2018), Cawthorne and Cenci (2019), Kumari et al. (2021), Iversen et al. (2020), Smits et al. (2022), Tokranova (2020), Kowe et al. (2022), Cruz-Martínez et al. (2021)
		Hope	Dadgar and Joshi (2015a,b), Smits et al. (2022)
Self-enhancement	Achievement (AC)	Capable	Umbrello and De Bellis (2018), Cawthorne and Robbins-van Wynsberghe (2019)
		Successful	Nouwen et al. (2015), Nouwen and Zaman (2018)
		Efficiency	Xu et al. (2012), Umbrello and De Bellis (2018), Umbrello (2019), Royakkers and Steen (2017), Maathuis et al. (2019), van de Kaa et al. (2020), Boyd (2022), Graubohm et al. (2020), Umbrello (2020)
		Competence	Van Wynsberghe (2013), Heger (2019), Van Wynsberghe (2012)
	Power (PO)	Preserving individuals' public image	Miller et al. (2007), Betz and Fritsch (2016), Friedman et al. (2006), Detweiler et al. (2010a)
		Social power	Xu et al. (2012), Borning et al. (2005), Rychwalska and Roszczynska-Kurasinska (2017)
		Recognition	Miller et al. (2007), Xu et al. (2012), Sapraz and Han (2021)
		Wealth	Simons and Verhagen (2008), Detweiler et al. (2010a)
		Ownership and property	Zaman and JafariNaimi (2015), Friedman and Kahn (2000), Le Dantec et al. (2009), Simons and Verhagen (2008), Borning et al. (2004), Friedman et al. (2006), van de Kaa et al. (2020), Grünloh (2018), Watkins (2018), Jaljolie et al. (2022), Graubohm et al. (2020), Burmeister et al. (2021), García and Gil (2020), Tokranova (2020)
		Knowledge	Le Dantec et al. (2009), Simons and Verhagen (2008), Detweiler et al. (2010a), Köhler et al. (2022)
	Hedonism (HE)	Pleasure	Helbing et al. (2021), Kowe et al. (2022)
		Calmness	van de Kaa et al. (2020), Watkins (2018), Cawthorne and Cenci (2019), Kumari et al. (2021), Iversen et al. (2020), Smits et al. (2022), Tokranova (2020), Kowe et al. (2022), Cruz-Martínez et al. (2021)
		Hope	Dadgar and Joshi (2015a,b), Smits et al. (2022)

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Table 3 (*continued*).

Schwartz's dimension	Schwartz's value category	Extracted values from the studies	Studies
Self-transcendence	Benevolence (*BE*)	Responsibility	Detweiler et al. (2010b), Le Dantec et al. (2009), Umbrello (2019), Borning et al. (2004), Van Wynsberghe (2013), Maathuis et al. (2019), Borning et al. (2005), Cummings (2006), Detweiler et al. (2010a), Friedman and Borning (2002), Rychwalska and Roszczynska-Kurasinska (2017), Cawthorne and Cenci (2019), Walton and DeRenzi (2009), Ast and Gaedke (2016), Detweiler and Hindriks (2012), Gan and Moussawi (2022), Lüthi et al. (2021), Van Wynsberghe (2012), Wambsganss et al. (2021), Helbing et al. (2021), Burmeister et al. (2021), García and Gil (2020), Sapraz and Han (2021), Tokranova (2020)
		Helpful	Harbers and Neerincx (2017, 2014), Köhler et al. (2022), Kowe et al. (2022)
		Trust	Miller et al. (2007), Betz and Fritsch (2016), Nouwen et al. (2015), Friedman and Kahn (2000), Briggs and Thomas (2015), Vermaas et al. (2010), Dadgar and Joshi (2015a), Maathuis et al. (2019), Cawthorne and Robbins-van Wynsberghe (2019), van de Kaa et al. (2020), Barn and Barn (2018), Dadgar and Joshi (2015b), Friedman et al. (2002), Borning et al. (2005), Cheon and Su (2016), Detweiler et al. (2010a), Nouwen and Zaman (2018), Grünloh (2018), Watkins (2018), Thornton et al. (2018), Heger (2019), Cawthorne and Cenci (2019), Harbers and Neerincx (2014), Walton and DeRenzi (2009), Detweiler and Hindriks (2012), Van Andel et al. (2015), Gan and Moussawi (2022), Lüthi et al. (2021), Oktay and Pender (2020), Wambsganss et al. (2021), Jaljolie et al. (2022), Graubohm et al. (2020), Dexe et al. (2020), Burmeister et al. (2021), García and Gil (2020), Sapraz and Han (2021), Köhler et al. (2022), Tokranova (2020), Kowe et al. (2022), Cruz-Martínez et al. (2021)
		Transparency	Walldius et al. (2005), Detweiler et al. (2010b), Umbrello (2019), Cawthorne and Robbins-van Wynsberghe (2019), Cheon and Su (2016), Detweiler et al. (2010a), Walldius et al. (2015), Thornton et al. (2018), Heger (2019), Borning et al. (2005), Ast and Gaedke (2016), Detweiler and Hindriks (2012), Sackey (2020), Vermaas et al. (2010), Friedman and Borning (2002), Harbers and Neerincx (2014), Rychwalska and Roszczynska-Kurasinska (2017), Gan and Moussawi (2022), Umbrello and Yampolskiy (2022), Wambsganss et al. (2021), Umbrello et al. (2021), Graubohm et al. (2020), Umbrello and Van de Poel (2021), Sapraz and Han (2021), Umbrello (2020)
		Usability	Alvarenga and Tanev (2017), Xu et al. (2012), Dadgar and Joshi (2015a), Maathuis et al. (2019), van de Kaa et al. (2020), Dadgar and Joshi (2015b), Heger (2019), Cawthorne and Cenci (2019), Boyd (2022), Jaljolie et al. (2022), Burmeister et al. (2021), García and Gil (2020), Sapraz and Han (2021), Umbrello (2020), Cenci and Cawthorne (2020), Tokranova (2020)
		Involvement	Miller et al. (2007), Nouwen et al. (2015), Le Dantec et al. (2009), Sambasivan and Jackson (2008), Badillo-Urquiola et al. (2020), Sackey (2020), Nouwen and Zaman (2018), Le Dantec and Poole (2008), Watkins (2018), Burrows et al. (2015), Oktay and Pender (2020), Dexe et al. (2020), Köhler et al. (2022), Cenci and Cawthorne (2020), Kowe et al. (2022)
		Togetherness	Alsheikh et al. (2011), Le Dantec et al. (2009), Maathuis et al. (2019), Gan and Moussawi (2022), Jaljolie et al. (2022), Köhler et al. (2022), Cenci and Cawthorne (2020)
		Integrity	Xu et al. (2012), Detweiler et al. (2010a), Jaljolie et al. (2022)
		Accuracy	Heger (2019)
		Continuity	Heger (2019)
	Universalism (*UN*)	Protecting the environment	Le Dantec et al. (2009), Borning et al. (2004), Cawthorne and Robbins-van Wynsberghe (2019), van de Kaa et al. (2020), Watkins (2018), Friedman and Borning (2002), Cawthorne and Cenci (2019), Gan and Moussawi (2022), Helbing et al. (2021), Graubohm et al. (2020), Iversen et al. (2020), Dexe et al. (2020), García and Gil (2020), Cenci and Cawthorne (2020), Tokranova (2020)
		Justice	Borning et al. (2004), van de Kaa et al. (2020), Vernim et al. (2022), Gan and Moussawi (2022), Jaljolie et al. (2022), Köhler et al. (2022)
		Equality	Detweiler et al. (2010b), Rychwalska and Roszczynska-Kurasinska (2017), Dexe et al. (2020), Cenci and Cawthorne (2020)
		Welfare	Walldius et al. (2005), Friedman and Kahn (2000), Friedman et al. (2006), Dadgar and Joshi (2015a), Cawthorne and Robbins-van Wynsberghe (2019), van de Kaa et al. (2020), Dadgar and Joshi (2015b), Cummings (2006), Walldius et al. (2015), Grünloh (2018), Watkins (2018), Heger (2019), Cawthorne and Cenci (2019), Harbers et al. (2010), Walton and DeRenzi (2009), Detweiler and Hindriks (2012), Vernim et al. (2022), Helbing et al. (2021), Jaljolie et al. (2022), Iversen et al. (2020), Smits et al. (2022), Burmeister et al. (2021), García and Gil (2020), Sapraz and Han (2021), Cenci and Cawthorne (2020), Tokranova (2020), Kowe et al. (2022), Cruz-Martínez et al. (2021), Dexe et al. (2020)
		Fairness	Walldius et al. (2005), Le Dantec et al. (2009), Borning et al. (2004), Cawthorne and Robbins-van Wynsberghe (2019), Borning et al. (2005), Detweiler et al. (2010a), Walldius et al. (2015), Watkins (2018), Thornton et al. (2018), Gan and Moussawi (2022), Wambsganss et al. (2021), Umbrello et al. (2021), Dexe et al. (2020), Umbrello and Van de Poel (2021), Sapraz and Han (2021), Köhler et al. (2022), Cenci and Cawthorne (2020)
		Freedom from bias	Zaman and JafariNaimi (2015), Le Dantec et al. (2009), Borning et al. (2005), Friedman and Borning (2002), Vernim et al. (2022), Lüthi et al. (2021), Tokranova (2020)
		Altruism	Sambasivan and Jackson (2008)

to the value categories of conformity (2 studies, 2.3%), tradition (2 studies, 2.3%), self-direction (12 studies, 14%), hedonism (2 studies, 2.3%), achievement (1 studies, 1%), power (2 studies, 2.3%), and universalism (13 studies, 15.3%). We describe them in detail in the next section.

4.3. RQ2: Ethical values

Human values are tacit in the design and development of software systems, and their identification is often overlooked.

The primary studies characterize and define human values in philosophical and social sciences' terms that we use for conceptualization.

There are different conceptualizations of values across the primary studies. For example, Heger (2019) refer to the value of anonymity in capturing data so that it cannot be associated with a specific person.

4.3.1. Identifying, conceptualizing, and categorizing values

We looked at how values are expressed and embedded in the software systems reported in the primary studies. As argued by Rescher (1969), consideration of different aspects of value categorizations can help understand further the concept of values. We adopted the Schwartz value structure, a classification widely used in disciplines like social sciences and ethics Schwartz (2012), Rosario et al. (2014), and categorized the values extracted from the primary studies[10] (see Fig. 2).

Table 3 provides an overview of the values identified in the primary studies and their classification based on this structure (see more details about the values, including explanations and examples in Tables 3–6 in online Appendix C[11]). Also, Table 4 summarizes the coverage by the primary studies of the value categories of Schwartz (2007, 1992), and Table 1 in online Appendix A[12] shows the categorized values and their codification at a finer detail.

In what follows, we provide a description of each value category covered in the primary studies as well as examples from the primary studies to elucidate how such categories of values can be accounted for in software systems. We summarize our findings with respect to each value category.

[10] In categorizing, we encountered values with multiple meanings which expressed motivational goals of more than one value category. To reduce the complexity of value relations and clarify the meaning of values in this structure, we only considered these values in one specific value category. This was possible by analyzing the precise meaning of values and their accordance with the primary goals of value categories.

[11] For the interested reader, Tables 3–6 are available here.

[12] For more details refer to Table 1 here.

Table 4
Coverage of the primary studies related to the value categories of Schwartz value structure. “SD” stands for self-direction, “ST” stimulation, “HE” hedonism, “AC” achievement, “PO” power, “SE” security, “CO” conformity, “TR” tradition, “BE” benevolence, and “UN” universalism.

Studies	SD	ST	HE	AC	PO	SE	CO	TR	BE	UN
Alsheikh et al. (2011), Burrows et al. (2015)	□	□	□	□	□	■	□	■	■	□
Miller et al. (2007), Betz and Fritsch (2016)	□	□	□	□	■	■	□	□	■	□
Alvarenga and Tanev (2017)	□	□	■	□	□	■	□	□	■	■
Yoo et al. (2013), Xu et al. (2008), Mithun et al. (2018)	□	□	□	□	□	■	□	□	□	□
Nouwen et al. (2015), Briggs and Thomas (2015), Vermaas et al. (2010), Sackey (2020), Harbers and Neerincx (2017), Ast and Gaedke (2016), Umbrello and Yampolskiy (2022)	□	□	□	□	□	■	□	□	■	□
Zaman and JafariNaimi (2015)	□	□	□	□	■	■	□	□	□	■
Xu et al. (2012)	□	□	■	■	■	■	□	□	■	■
Umbrello and De Bellis (2018), Royakkers and Steen (2017)	□	□	□	■	□	■	□	□	□	□
Le Dantec et al. (2009), Friedman et al. (2006), Grünloh (2018), Jaljolie et al. (2022), Burmeister et al. (2021), García and Gil (2020), Sapraz and Han (2021)	■	□	□	□	■	■	□	□	■	■
Sambasivan and Jackson (2008), Umbrello and Van de Poel (2021), Cenci and Cawthorne (2020), Umbrello et al. (2021)	■	□	□	□	□	□	□	□	■	■
Simons and Verhagen (2008), Denning et al. (2014)	□	□	□	□	■	■	□	□	□	□
Umbrello (2019, 2020)	□	□	□	■	□	■	□	□	■	□
Badillo-Urquiola et al. (2020), Barn and Barn (2018), Friedman et al. (2002), Oktay and Pender (2020)	■	□	□	□	□	■	□	□	■	□
Borning et al. (2004)	□	□	□	□	■	□	□	□	■	■
Van Wynsberghe (2013, 2012)	■	□	□	■	□	□	□	□	■	□
Maathuis et al. (2019)	■	□	■	■	■	■	□	□	■	■
Cawthorne and Robbins-van Wynsberghe (2019)	□	□	□	■	□	■	□	□	■	■
van de Kaa et al. (2020)	■	□	■	■	■	■	■	□	■	■
Walldius et al. (2005), Detweiler et al. (2010b), Cummings (2006), Walldius et al. (2015), Friedman and Borning (2002), Dadgar and Joshi (2015a,b), Cawthorne and Cenci (2019)	■	□	■	□	□	■	□	□	■	■
Friedman and Kahn (2000), Borning et al. (2005), Detweiler et al. (2010a)	□	□	□	□	■	■	□	□	■	■
Cheon and Su (2016)	■	□	□	□	□	■	□	■	■	□
Nouwen and Zaman (2018)	■	□	□	■	□	■	□	□	■	□
Le Dantec and Poole (2008)	■	□	□	□	□	□	□	□	■	□
Ghosh (2016), Primiero et al. (2020)	■	□	□	□	□	■	□	□	□	□
Watkins (2018)	■	□	■	□	■	■	■	□	■	■
Thornton et al. (2018)	■	□	□	□	□	■	□	■	■	■
Rychwalska and Roszczynska-Kurasinska (2017), Graubohm et al. (2020)	■	□	□	■	■	■	■	□	■	■
Heger (2019)	■	□	■	■	□	■	□	□	■	■
Harbers and Neerincx (2014)	■	□	□	□	■	■	■	□	■	□
Harbers et al. (2010)	□	□	□	□	□	□	□	□	■	■
Walton and DeRenzi (2009), Köhler et al. (2022)	■	□	□	□	■	■	■	□	■	■
Detweiler and Hindriks (2012), Van Andel et al. (2015), Gan and Moussawi (2022)	■	□	□	□	□	■	□	□	■	■
Vernim et al. (2022)	■	□	□	□	□	□	□	□	□	■
Lüthi et al. (2021)	■	□	□	□	□	■	■	□	■	■
Boyd (2022)	□	□	□	■	□	□	■	□	■	□
Kumari et al. (2021)	□	□	■	□	□	■	□	□	□	□
Wambsganss et al. (2021), Dexe et al. (2020)	□	□	□	□	□	■	□	□	■	■
Helbing et al. (2021)	□	□	■	□	□	□	□	□	■	■
Smits et al. (2022)	■	□	■	□	□	■	□	□	□	■
Tokranova (2020)	■	□	■	□	■	■	□	□	■	■
Kowe et al. (2022)	■	□	■	□	□	■	■	□	■	■
Cruz-Martínez et al. (2021)	■	□	■	□	□	■	□	■	■	■
Iversen et al. (2020)	□	□	■	□	□	■	□	□	□	■

- *Security* is about caring for and keeping individuals and data safe against potential harm. Security features protect individuals' health (both physically and mentally), their personal information, identity, and safety against unintended hazards, physical risks, and threats. They also concern with controlling and securing data and keeping their flow secure against system failure and data loss. Consider the case of software enabled by an affective car assistance system for traffic safety. This software captures the driver's face to calculate the driver's emotional state (like anger, happiness, or fear) and gives situation-specific warnings. Security in this example can be incorporated by protecting the individuals' data and their *anonymity* by not using cloud-based solutions and personal data that are unnecessary for the system's functioning and also concealing all information about the driver (Heger, 2019). Security has been studied in the context of software systems in 64 out of 85 studies (see Table 4). The most prevalent values in this category are *privacy* and *safety* (see Table 3).

Table 5
Methods and techniques to elicit values from stakeholders in the primary studies.

Methods and techniques	Example	Studies
Interviews	Through interviews in Miller et al. (2007), the participants indicated their views on values-related issues and how those issues can be handled.	Miller et al. (2007), Detweiler et al. (2010a,b), Sambasivan and Jackson (2008), Borning et al. (2005), Xu et al. (2012), Nouwen and Zaman (2018), Heger (2019), Nouwen et al. (2015), Zaman and JafariNaimi (2015), Denning et al. (2014), Briggs and Thomas (2015), Harbers and Neerincx (2017), Maathuis et al. (2019), Barn and Barn (2018), Walldius et al. (2005), Alsheikh et al. (2011), Friedman et al. (2002), Royakkers and Steen (2017), Walton and DeRenzi (2009), van de Kaa et al. (2020), Lüthi et al. (2021), Boyd (2022), Kumari et al. (2021), Wambsganss et al. (2021), Jaljolie et al. (2022), Smits et al. (2022), Sapraz and Han (2021), Köhler et al. (2022), Iqbal et al. (2021), Kowe et al. (2022), Graubohm et al. (2020)
Workshops	Through a workshop in Nouwen and Zaman (2018), Nouwen et al. (2015), Zaman and JafariNaimi (2015), some personas were created to help participants to become familiar with the subject (*i.e.,* parental mediation). Then according to the personas and the extracted values from the stakeholders, the other stakeholder group (*i.e.,* system designers) was asked to build a value hierarchy to make value alignment and discuss the conflicting stakeholder values.	Nouwen and Zaman (2018), Nouwen et al. (2015), Zaman and JafariNaimi (2015), Denning et al. (2014), Briggs and Thomas (2015), Harbers and Neerincx (2014, 2017), Maathuis et al. (2019), Barn and Barn (2018), Royakkers and Steen (2017), Sackey (2020), Oktay and Pender (2020), Dexe et al. (2020), Primiero et al. (2020)
Case studies	In a case study in Detweiler et al. (2010a), the participants' values were captured, and the extent to which values could be addressed in the system (in accordance with ethical principles) was discovered.	Detweiler et al. (2010a,b), Heger (2019), Zaman and JafariNaimi (2015), Royakkers and Steen (2017), Ghosh (2016), Le Dantec et al. (2009), Le Dantec and Poole (2008), Cawthorne and Cenci (2019), Cummings (2006), Cawthorne and Robbins-van Wynsberghe (2019), Grünloh (2018), Umbrello (2019)
Mock-ups and prototypes	In a co-design activity in Yoo et al. (2013), the participants developed some prototypes for the phone to express their ideas for using mobile phones to provide safety for homeless young people.	Borning et al. (2005), Xu et al. (2012), Denning et al. (2014), Harbers and Neerincx (2017), Maathuis et al. (2019), Barn and Barn (2018), Friedman et al. (2002), Yoo et al. (2013), Cawthorne and Cenci (2019), Cummings (2006), Oktay and Pender (2020), Umbrello et al. (2021)
Surveys	Through a survey in Van Andel et al. (2015), the participants indicated their interpretations, views, and comments about the relevant values of the system and rated the values in order of their importance.	Miller et al. (2007), Mithun et al. (2018), Van Andel et al. (2015), Barn and Barn (2018), Walldius et al. (2005), Ghosh (2016), Walldius et al. (2015), Kumari et al. (2021), Tokranova (2020), Cruz-Martínez et al. (2021)
Focus groups	In a focus group in Maathuis et al. (2019), the participants discussed their experience with the use of the system (*i.e.,* web-based technologies) in their professional lives and the relevant ethical values in the design of the system.	Heger (2019), Harbers and Neerincx (2017), Maathuis et al. (2019), Burrows et al. (2015), Barn and Barn (2018), Walton and DeRenzi (2009), Kumari et al. (2021), Jaljolie et al. (2022)
Sketches	The value sketches in Yoo et al. (2013) were about the understanding of participants on mobile phones and safety. The sketches represented the participants' perceptions about where and when homeless young people might feel unsafe.	Badillo-Urquiola et al. (2020), Yoo et al. (2013)
Value scenarios and story-boards	The presented scenarios in Harbers and Neerincx (2017) aimed to explain the functioning and envision the values of the virtual assistant in a train traffic control context for different operators.	Harbers and Neerincx (2017), Yoo et al. (2013)
Photo-elicitation	The photo-elicitation in Le Dantec et al. (2009), Le Dantec and Poole (2008) allowed an understanding of the perceptions and use of technology by the homeless. Using this method, homeless people could express their values concerning technology, its use, and its relationship to their daily lives in the homeless community.	Le Dantec et al. (2009), Le Dantec and Poole (2008)

- *Conformity* is about adapting the systems to meet the stakeholders' needs. Consider the case of software enabled by a smart grid system (called smart metering) that records electricity, gas, or water consumption and communicates that information for monitoring and billing. Conformity in this example can be accounted for by modifying generation or consumption patterns in reaction to an external signal like a price change to meet the consumers' needs, *e.g.,* using water at an affordable cost (van de Kaa et al., 2020). Six primary studies reported the value of *flexibility* in the conformity value category (see Table 3).
- *Tradition* is about caring to respect the behaviors, culture, and habits of individuals and groups in the social environment. Consider the case of communication media in an Arabic cultural context. The media can provide interactions between men and women with respect to certain conventions. For example, by providing some features such as friend suggestions that can prioritize females as friends for women or using kin relationships to suggest appropriate male friends, the media technologies care for *cultural values* (in this case, Islamic values) (Alsheikh et al., 2011). There are 4 primary studies expressed values such as *cultural values* and *lifestyle values* that belong to tradition value category (see Table 3).
- *Self-direction* is about caring for the rights of individuals in determining how they should be treated; have autonomy in their choices and control over activities. Consider the case of pervasive software for elderly care in a residential setting. This software provides various services for elders at home like queries, alarms, detecting falls, recognizing specific behaviors, and monitoring the health status. By allowing the elderly to stay at home as they get older (*i.e.,* aging-in-place) without depending on their families, this software supports the *autonomy* and the *independence* (Detweiler and Hindriks, 2012) and allows *self-direction*. Self-direction is one of the most recurring value categories expressed and incorporated in the primary studies, mentioned in 45 out of 85 studies (see Table 4).
- *Hedonism* can be incorporated into software systems by caring for pleasure or sensual gratification when using the system. Consider the case of healthcare drones software that allows delivery of blood donations, vaccinations, medications, and other medical

supplies (Cawthorne and Cenci, 2019; Cawthorne and Robbins-van Wynsberghe, 2019). Hedonism, in this example, can be incorporated by enhancing the *calmness* of local communities and authorities by, for instance, replacing loud and disruptive medical helicopters. Twelve (12) out of 85 studies expressed values such as *pleasure*, *calmness*, and *hope* that belong to hedonism value category (see Table 3).

- *Achievement* can be embedded in a software system by supporting users to carry out the activities successfully, leading to an increase in individuals' personal satisfaction. Consider the case of software enabled by a sensor-based physiotherapeutic assistance system, used for the patients' treatment in a home setting (Heger, 2019). Achievement, in this example, can be accounted for by offering physiotherapeutic exercises at home through live video coaching via the software system. As shown in Table 3, we found four values belonging to achievement category: *capable*, *successful*, *efficiency*, and *competence*. In total, 15 primary studies reported values in this category.
- *Power* can be incorporated into software systems by caring for dominance over people and resources, and by maintaining social status. Consider the case of an EHR system. By allowing patients to control their information (*i.e.,* resource), an EHR system cares for the value of the *ownership* and *property*, *i.e.,* the patient has the ability to manage and own their health information (Grünloh, 2018). In total, 22 primary studies reported values that we mapped to the power value category (see Table 4).
- *Benevolence* can be incorporated into software systems by being kind, like helping individuals to do their tasks. Consider the case of software enabled by a virtual assistant that supports workload harmonization in the context of train traffic control. By sharing workload information to support operators to help each other when needed and yielding a more even distribution of workload over the team members, this software cares for the value of *helpful* (Harbers and Neerincx, 2017). For instance, if a virtual assistant detects a high and low workload level for operators A and B, respectively, sharing this information enables operator B to provide timely assistance to operator A in task completion. We found values like *usability* and *transparency* that belong to the benevolence category but are not part of Schwartz value structure (see Table 3). The focus of these values on the helpfulness of the system is a common aspect among them. We also found benevolence is one of the most recurring value categories expressed and incorporated in the primary studies, mentioned in 68 out of 85 studies (see Table 4).
- *Universalism* is about preserving the welfare of people and protecting the environment through software systems (*e.g.,* safeguarding the individuals' physical and mental health, overcoming unfairness perpetrated on individuals). Consider the case of software to simulate and predict urban development patterns over time. For instance, by not discriminating unfairly against any group of individuals to utilize specific infrastructural facilities like transport and communication, this software cares for the value of *freedom from bias* (Borning et al., 2004). We found that universalism is one of the most recurring value categories, expressed in 46 out of 85 primary studies (see Table 4).

Finally, Table 4 shows that the value category of *stimulation*, involving values such as *excitement* and *novelty*, is not addressed by any of the primary studies.

4.3.2. Eliciting values from stakeholders

Values are expressions of what humans and organizations consider worthy but extracting them is difficult because they are not measurable and they differ with changing perspectives. In our analysis, we elicited methods and techniques used in the primary studies to support extraction of values and reflection of ethical issues. Extraction methods and techniques,together with examples from the primary studies, are presented in Table 5. Some studies used a mixed-method approach in the design process, *e.g.,* methods like sketches and prototypes were used in combination with other methods like interviews.

Methods and techniques to elicit values from the stakeholders. According to the primary studies, we found various methods and techniques to extract values from the stakeholders in different situations.

Interviews. Interviews were mainly used to collect individuals' opinions, views, and values to understand their judgments about a use context,[13] an existing technology, or a proposed design.

Surveys. Surveys were used to extract views and values from the individuals, measure individuals' needs, solicit their views on values, discover value-related issues and their implications, and rank values.

Workshops. Workshops allowed the individuals to collaboratively discuss, analyze, and share the value-related issues and ethical considerations for a specific system. In particular, workshops allowed the individuals to identify values and value tensions, expose ethical issues, and recognize the value implications.

Case studies. Case studies were used to examine a real-life situation. The focus was to have a zoomed-in detailed look at real-world complexities like how ethical values can be satisfied in the system or to what extent ethical considerations can be considered during software design. Case studies could help assess the strengths and challenges of appropriate methods like the value sensitive design.

Focus groups. Focus groups allowed the individuals to cooperatively express their opinions and attitudes in discussions about the key values, value-related issues, and implications. In contrast to workshops, focus groups were not used to develop a consensus for a decision among all participants. Focus groups concentrated on a specific topic or issue relevant to the discussion.

Sketches. Sketches were mainly used to provide non-verbal understandings of a system using visual expressions and focused on the participants' views and values. In particular, sketches assisted the individuals in making value-related decisions by understanding what is important for them in relation to a system, how a system is situated in a context, and how values are implicated by system functioning.

Mock-ups and prototypes. Mock-ups and prototypes were used to make software designs more concrete for the individuals and provide a visual way to refer to systems. They allowed the participants to investigate the system's functions, interactions among stakeholders and system, implications of systems on stakeholders, value tensions, and the system's suitability in a specific context.

Value scenarios and story-boards. Value scenarios and story-boards were used to help the individuals discuss and communicate about different system features, especially humanistic and societal considerations. In particular, value scenarios and story-boards assisted participants in focusing on implications for stakeholders, key values, use contexts, and longer-term and potential impacts.

Photo-elicitation. Photo-elicitation was used to elicit comments and views from the individuals. Visual images (such as photographs and paintings) allowed the participants to direct the discussion about a system and enabled them to speak in greater detail about their situation, concerns, and key values from their perspective.

4.3.3. Relationships among values

Values have relationships among them (Schwartz, 2012). They can conflict or reinforce one another. For example, consider a conflict between two values of *well-being* and *autonomy* in a sensor-based assistance system that supports physio-therapeutic activities at

[13] It is the conditions (including the technical, social, and organizational environments) under which the intended users utilize an artifact (like software products) to achieve their desired goals.

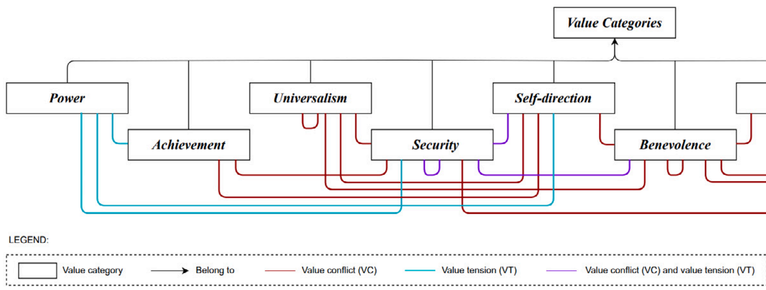

Fig. 6. The value conflicts (VCs) and value tensions (VTs) among the value categories.

home (Heger, 2019). This system that aims to support the well-being of patients can conflict with the value of autonomy since this value requires the system to support the ideal ways of executing the treatment for the patients.

Value conflicts and value tensions. In our study, we found two types of relationships: *value conflict* and *value tension*. Value conflict refers to situations in which technology supports one value while undermines another one. Value tension, in turn, refers to instances in which two values are in conflict in a given situation (Friedman and Hendry, 2012). Although value conflict and value tension theoretically are different concepts, *i.e.,* value tension is a general term mentioning a value conflict; they were used quite similarly in the primary studies. Most studies generally discussed these two concepts without mentioning any example: 20 out of 85 studies explained them with example(s) (see Table 6).

Fig. 6 depicts the representation of value conflicts and value tensions within the corresponding value categories, as inferred from the identified relationships among the values in this table. According to the figure, the categories that have the highest level of conflict and/or tension with other categories are *security*– contradicts with seven categories (*i.e.,* benevolence, self-direction, universalism, achievement, power, security, and conformity) and *benevolence*– contradicts with seven categories (*i.e.,* benevolence, security, universalism, tradition, hedonism, self-direction, and conformity). Moreover, the figure shows that specific categories exclusively exhibit either value conflict (depicted in red) or value tension (depicted in blue). For example, the category of achievement predominantly exhibits conflict relationships with other categories. However, there are some categories that exhibit both types of relationships (depicted in purple). One such example is security, which experiences both conflict and tension, as seen in its interactions with categories like self-direction. Further, we noticed that stimulation does not have any relation with other categories.

Privacy, autonomy, safety, and involvement are the most conflicting values. Values can be conflicting in the same category, or they can be conflicting with another value category, see Table 6. An example of the former case, the safety of a person may conflict with the safety of society. An example of cross-value conflict is privacy of a suspect may conflict with welfare of society. We found that the most conflicting values are: (i) *privacy* conflicts with fourteen values, (ii) *autonomy* conflicts with nine values, (iii) *safety* conflicts with seven values, (iv) *involvement* conflicts with six values.

Resolving value conflicts and tensions. To resolve the value conflicts and value tensions in software design, a few primary studies (Miller et al., 2007; Betz and Fritsch, 2016; Nouwen et al., 2015; Zaman and JafariNaimi, 2015; Nouwen and Zaman, 2018; Lüthi et al., 2021) focused on addressing conflicts and tensions among the values and proposed the methods of *value dams and value flows* and *value alignment*.

Value dams and value flows[14] by keeping the design options supporting ethical values and ignoring those undermining values, helps to resolve the value tensions. For example, consider the use of this method for resolving value tensions among *privacy* and *awareness* in Miller et al. (2007). In this example, value dams are design options like *logging searches* and *logging queries* which concern the value of privacy, and many stakeholders agreed that these options could violate their privacy. Value flows are design options such as *reporting the frequency of use of the participants' codes* and *showing the ranking of participants' posts by their peers* which are related to the value of awareness. A majority of stakeholders agreed that they would like to have these two options. Thus, to reduce the privacy-related value dams and benefit from the awareness-related value flows, it was decided not to log or report who searches or queries but to log and report the frequency of code use and implement content ranking.

Value alignment[15] by prioritizing and aligning the values of different stakeholder groups helps to make trade-offs among the values. For example, in Nouwen et al. (2015), there are two stakeholder groups: media company employees and children's parents. The media company employees built a hierarchy of their values in relation to the media without considering the parents' values. Then, the employees tried to map the parents' values onto their values and find the value conflicts.

4.4. RQ3: Embedding stakeholders' requirements and values in software design and development

VSD is one of the comprehensive and common methods to account for human values in software design (Friedman et al., 2013; Winkler and Spiekermann, 2018). We used it as a basis to analyze the primary studies (see Fig. 1). Table 7 shows three investigations and conducted activities of VSD in our primary studies, and Table 8 shows the coverage of the conducted activities in these studies.

Most studies focus on the identification of stakeholders, ethical values, and the benefits and harms from the systems. According to Tables 7 and 8, among all undertaken activities in three investigations of VSD (*i.e.,* conceptual, empirical, and technical), $C1$, $C2$, and $C3$ occur in the primary studies more frequently than other activities.

A few studies focus on prioritizing values and translating them into software requirements. Based on Tables 7 and 8, a few studies refer to activities $E2$ and $T1$ (2/85 and 4/85 studies, respectively). It means they put less effort into prioritizing values and translating them into more concrete[16] constructs (*e.g.,* software requirements).

None of the primary studies covers all the depicted activities in regard to embedding values in software design and development. Based on Tables 7 and 8, none of the primary studies focuses on all the activities to embed the values in software design and development.

[14] In this method, when a small percentage of stakeholders strongly object to a design option, it could be removed from the design space (known as the value dams). Also, when a good percentage of stakeholders find a design option appealing, it could remain in the design space (known as the value flows).

[15] In this method, one stakeholder group should prioritize the values based on their goals in the system without knowing the others' values. Afterward, they should align the other stakeholder groups' values to their values through discussions and find the value conflicts.

[16] It refers to making the values simple and clear enough (instead of abstract) to allow design decision makers to look at them and figure out what they are.

Table 6
Value conflicts and value tensions extracted from the primary studies.

Extracted value conflicts (VCs) and value tensions (VTs) from the studies	Value conflicts (VCs) and value tensions (VTs) among corresponding value categories	Studies
• **Privacy (VCs):** Privacy vs. involvement, privacy vs. responsibility, privacy vs. transparency, privacy vs. welfare, privacy vs. creativity, privacy vs. informed consent, privacy vs. usability, privacy vs. safety, privacy vs. support and protection, privacy vs. togetherness • **Privacy (VTs):** Privacy vs. reputation, privacy vs. trust, privacy vs. safety, privacy vs. autonomy, privacy vs. social recognition, privacy vs. transparency, privacy vs. responsibility • **Safety (VCs):** Safety vs. efficiency, safety vs. welfare, safety vs. freedom, safety vs. flexibility, safety vs. privacy • **Safety (VTs):** Safety vs. autonomy, safety vs. privacy, safety vs. involvement • **Certainty (VCs):** Certainty vs. competence • **Control (VCs):** Control vs. involvement • **Control (VTs):** Control vs. autonomy • **Availability (VCs):** Availability vs. equality • **Anonymity (VCs):** Anonymity vs. responsibility, anonymity vs. transparency, anonymity vs. identity	• **Security (VCs):** Security vs. benevolence, security vs. universalism, security vs. self-direction, security vs. security, security vs. achievement, security vs. conformity • **Security (VTs):** Security vs. power, security vs. benevolence, security vs. security, security vs. self-direction	Miller et al. (2007), Betz and Fritsch (2016), Simons and Verhagen (2008), Badillo-Urquiola et al. (2020), Maathuis et al. (2019), Detweiler et al. (2010a), Nouwen and Zaman (2018), Watkins (2018), Thornton et al. (2018), Heger (2019), Cawthorne and Cenci (2019), Umbrello and Yampolskiy (2022), Jaljolie et al. (2022), García and Gil (2020), Umbrello (2020), Graubohm et al. (2020), Köhler et al. (2022)
• **Accuracy (VCs):** Accuracy vs. dignity • **Involvement (VCs):** Involvement vs. cultural values, involvement vs. privacy, involvement vs. control, involvement vs. flexibility, involvement vs. identity • **Involvement (VTs):** Involvement vs. safety • **Transparency (VCs):** Transparency vs. privacy, transparency vs. anonymity • **Trust (VCs):** Trust vs. responsibility, trust vs. anonymity, trust vs. calmness, trust vs. support and protection, trust vs. togetherness • **Trust (VTs):** Trust vs. privacy	• **Benevolence (VCs):** Benevolence vs. self-direction, benevolence vs. tradition, benevolence vs. security, benevolence vs. conformity, benevolence vs. benevolence, benevolence vs. hedonism, benevolence vs. universalism • **Benevolence (VTs):** Benevolence vs. security	Alsheikh et al. (2011), Betz and Fritsch (2016), Simons and Verhagen (2008), Maathuis et al. (2019), Detweiler et al. (2010a), Nouwen and Zaman (2018), Heger (2019), Sackey (2020), Jaljolie et al. (2022), Köhler et al. (2022)
• **Autonomy (VCs):** Autonomy vs. community, autonomy vs. welfare, autonomy vs. safety, autonomy vs. helpful, autonomy vs. support and protection, autonomy vs. togetherness • **Autonomy (VTs):** Autonomy vs. safety, autonomy vs. control, autonomy vs. privacy, autonomy vs. power • **Creativity (VCs):** Creativity vs. privacy • **Dignity (VCs):** Dignity vs. accuracy • **Identity (VCs):** Identity vs. anonymity	• **Self-direction (VCs):** Self-direction vs. universalism, self-direction vs. security, self-direction vs. benevolence, self-direction vs. achievement • **Self-direction (VTs):** Self-direction vs. security, self-direction vs. power	Badillo-Urquiola et al. (2020), Maathuis et al. (2019), Detweiler et al. (2010a), Watkins (2018), Heger (2019), Köhler et al. (2022)
• **Efficiency (VCs):** Efficiency vs. safety, efficiency vs. freedom • **Efficiency (VTs):** Efficiency vs. power	• **Achievement (VCs):** Achievement vs. security, achievement vs. self-direction • **Achievement (VTs):** Achievement vs. power	Maathuis et al. (2019), Watkins (2018), Thornton et al. (2018), Helbing et al. (2021)
• **Welfare (VCs):** Welfare vs. protecting the environment, welfare vs. safety, welfare vs. privacy, welfare vs. autonomy • **Justice (VCs):** Justice vs. support and protection, justice vs. togetherness	• **Universalism (VCs):** Universalism vs. universalism, universalism vs. security, universalism vs. self-direction, universalism vs. benevolence	Cawthorne and Cenci (2019), Heger (2019), Köhler et al. (2022)

Table 7
An overview of the conducted activities in the primary studies to embed the (ethical) values in design and development of software systems which is based on VSD methods (Fig. 1).

VSD investigations	Activities	Description	Studies
Conceptual investigations	C1	Identifying stakeholders (explicitly)	Miller et al. (2007), Betz and Fritsch (2016), Alvarenga and Tanev (2017), Walldius et al. (2005), Nouwen et al. (2015), Zaman and JafariNaimi (2015), Le Dantec et al. (2009), Xu et al. (2008), Briggs and Thomas (2015), Simons and Verhagen (2008), Denning et al. (2014), Borning et al. (2004), Vermaas et al. (2010), Royakkers and Steen (2017), Maathuis et al. (2019), Walldius et al. (2015), Mithun et al. (2018), Le Dantec and Poole (2008), Grünloh (2018), Thornton et al. (2018), Harbers and Neerincx (2014), Walton and DeRenzi (2009), Detweiler and Hindriks (2012), Vernim et al. (2022), Lüthi et al. (2021), Umbrello and Yampolskiy (2022), Boyd (2022), Kumari et al. (2021), Helbing et al. (2021), Jaljolie et al. (2022), Smits et al. (2022), García and Gil (2020), Sapraz and Han (2021), Köhler et al. (2022), Cenci and Cawthorne (2020), Tokranova (2020), Kowe et al. (2022), Cruz-Martínez et al. (2021), Dexe et al. (2020), Iversen et al. (2020), Graubohm et al. (2020)
		Identifying stakeholders (implicitly)	Alsheikh et al. (2011), Yoo et al. (2013), Xu et al. (2012), Detweiler et al. (2010b), Umbrello and De Bellis (2018), Sambasivan and Jackson (2008), Umbrello (2019), Badillo-Urquiola et al. (2020), Friedman et al. (2006), Dadgar and Joshi (2015a), Cawthorne and Robbins-van Wynsberghe (2019), van de Kaa et al. (2020), Barn and Barn (2018), Dadgar and Joshi (2015b), Friedman et al. (2002), Borning et al. (2005), Cheon and Su (2016), Sackey (2020), Detweiler et al. (2010a), Nouwen and Zaman (2018), Ghosh (2016), Watkins (2018), Burrows et al. (2015), Friedman and Borning (2002), Rychwalska and Roszczynska-Kurasinska (2017), Heger (2019), Cawthorne and Cenci (2019), Harbers and Neerincx (2017), Harbers et al. (2010), Van Andel et al. (2015), Umbrello and Van de Poel (2021), Umbrello (2020)
	C2	Identifying benefits and harms imposed by the system on stakeholder groups (explicitly)	Miller et al. (2007), Detweiler et al. (2010b), Umbrello and De Bellis (2018), Cawthorne and Robbins-van Wynsberghe (2019), Detweiler et al. (2010a), Rychwalska and Roszczynska-Kurasinska (2017), Cawthorne and Cenci (2019), Harbers and Neerincx (2014), Detweiler and Hindriks (2012), Van Andel et al. (2015), Lüthi et al. (2021), Umbrello and Yampolskiy (2022), Boyd (2022), Kumari et al. (2021), Jaljolie et al. (2022), Smits et al. (2022), García and Gil (2020), Sapraz and Han (2021), Köhler et al. (2022), Cenci and Cawthorne (2020), Kowe et al. (2022), Cruz-Martínez et al. (2021), Iversen et al. (2020)
		Identifying benefits and harms imposed by the system on stakeholder groups (implicitly)	Alsheikh et al. (2011), Betz and Fritsch (2016), Alvarenga and Tanev (2017), Walldius et al. (2005), Yoo et al. (2013), Nouwen et al. (2015), Zaman and JafariNaimi (2015), Xu et al. (2012), Friedman and Kahn (2000), Le Dantec et al. (2009), Xu et al. (2008), Briggs and Thomas (2015), Sambasivan and Jackson (2008), Simons and Verhagen (2008), Umbrello (2019), Badillo-Urquiola et al. (2020), Denning et al. (2014), Borning et al. (2004), Vermaas et al. (2010), Van Wynsberghe (2013), Royakkers and Steen (2017), Friedman et al. (2006), Dadgar and Joshi (2015a), Maathuis et al. (2019), van de Kaa et al. (2020), Barn and Barn (2018), Dadgar and Joshi (2015b), Friedman et al. (2002), Borning et al. (2005), Cummings (2006), Cheon and Su (2016), Sackey (2020), Nouwen and Zaman (2018), Walldius et al. (2015), Mithun et al. (2018), Le Dantec and Poole (2008), Ghosh (2016), Grünloh (2018), Watkins (2018), Burrows et al. (2015), Friedman and Borning (2002), Thornton et al. (2018), Heger (2019), Harbers and Neerincx (2017), Harbers et al. (2010), Walton and DeRenzi (2009), Ast and Gaedke (2016), Vernim et al. (2022), Umbrello and Van de Poel (2021)
	C3	Mapping benefits and harms onto corresponding values	Alsheikh et al. (2011), Miller et al. (2007), Betz and Fritsch (2016), Alvarenga and Tanev (2017), Walldius et al. (2005), Yoo et al. (2013), Nouwen et al. (2015), Zaman and JafariNaimi (2015), Xu et al. (2012), Detweiler et al. (2010b), Friedman and Kahn (2000), Umbrello and De Bellis (2018), Le Dantec et al. (2009), Xu et al. (2008), Briggs and Thomas (2015), Sambasivan and Jackson (2008), Simons and Verhagen (2008), Umbrello (2019), Badillo-Urquiola et al. (2020), Denning et al. (2014), Borning et al. (2004), Vermaas et al. (2010), Van Wynsberghe (2013), Royakkers and Steen (2017), Friedman et al. (2006), Dadgar and Joshi (2015a), Maathuis et al. (2019), Cawthorne and Robbins-van Wynsberghe (2019), van de Kaa et al. (2020), Barn and Barn (2018), Dadgar and Joshi (2015b), Friedman et al. (2002), Borning et al. (2005), Cummings (2006), Cheon and Su (2016), Sackey (2020), Detweiler et al. (2010a), Nouwen and Zaman (2018), Walldius et al. (2015), Mithun et al. (2018), Le Dantec and Poole (2008), Ghosh (2016), Grünloh (2018), Watkins (2018), Burrows et al. (2015), Friedman and Borning (2002), Thornton et al. (2018), Rychwalska and Roszczynska-Kurasinska (2017), Heger (2019), Cawthorne and Cenci (2019), Harbers and Neerincx (2017, 2014), Harbers et al. (2010), Walton and DeRenzi (2009), Ast and Gaedke (2016), Detweiler and Hindriks (2012), Van Andel et al. (2015), Vernim et al. (2022), Gan and Moussawi (2022), Lüthi et al. (2021), Umbrello and Yampolskiy (2022), Van Wynsberghe (2012), Boyd (2022), Kumari et al. (2021), Wambsganss et al. (2021), Helbing et al. (2021), Jaljolie et al. (2022), Smits et al. (2022), Burmeister et al. (2021), García and Gil (2020), Sapraz and Han (2021), Umbrello and Van de Poel (2021), Köhler et al. (2022), Umbrello (2020), Cenci and Cawthorne (2020), Tokranova (2020), Kowe et al. (2022), Cruz-Martínez et al. (2021), Dexe et al. (2020), Iversen et al. (2020), Graubohm et al. (2020), Umbrello et al. (2021), Primiero et al. (2020)
	C4	Categorizing the identified values	Walldius et al. (2005), Borning et al. (2004, 2005), Walldius et al. (2015), Friedman and Borning (2002), Ast and Gaedke (2016)
	C5	Identifying potential value tensions and conflicts	Alsheikh et al. (2011), Miller et al. (2007), Betz and Fritsch (2016), Nouwen et al. (2015), Zaman and JafariNaimi (2015), Umbrello and De Bellis (2018), Simons and Verhagen (2008), Umbrello (2019), Badillo-Urquiola et al. (2020), Maathuis et al. (2019), Sackey (2020), Detweiler et al. (2010a), Nouwen and Zaman (2018), Watkins (2018), Thornton et al. (2018), Rychwalska and Roszczynska-Kurasinska (2017), Heger (2019), Cawthorne and Cenci (2019), Umbrello and Yampolskiy (2022), Jaljolie et al. (2022), Köhler et al. (2022), Umbrello (2020), Graubohm et al. (2020)
Empirical investigations	E1	Understanding how stakeholders experience a technology with regard to the values they consider important	Alsheikh et al. (2011), Miller et al. (2007), Yoo et al. (2013), Nouwen et al. (2015), Zaman and JafariNaimi (2015), Xu et al. (2012), Detweiler et al. (2010b), Le Dantec et al. (2009), Briggs and Thomas (2015), Sambasivan and Jackson (2008), Badillo-Urquiola et al. (2020), Denning et al. (2014), Royakkers and Steen (2017), Dadgar and Joshi (2015a), Maathuis et al. (2019), van de Kaa et al. (2020), Barn and Barn (2018), Borning et al. (2005), Cheon and Su (2016), Detweiler et al. (2010a), Nouwen and Zaman (2018), Mithun et al. (2018), Le Dantec and Poole (2008), Ghosh (2016), Burrows et al. (2015), Heger (2019), Harbers and Neerincx (2017, 2014), Van Andel et al. (2015), Lüthi et al. (2021), Oktay and Pender (2020), Boyd (2022), Kumari et al. (2021), Wambsganss et al. (2021), Jaljolie et al. (2022), Smits et al. (2022), García and Gil (2020), Sapraz and Han (2021), Köhler et al. (2022), Kowe et al. (2022), Cruz-Martínez et al. (2021), Dexe et al. (2020), Graubohm et al. (2020), Primiero et al. (2020)
	E2	Prioritizing the values	Van Andel et al. (2015), Tokranova (2020)
	E3	Resolving value tensions and conflicts through making trade-offs among competing values	Miller et al. (2007), Betz and Fritsch (2016), Nouwen et al. (2015), Zaman and JafariNaimi (2015), Nouwen and Zaman (2018), Lüthi et al. (2021)
Technical investigations	T1	Translating the identified values into software requirements	Umbrello (2019), Cawthorne and Robbins-van Wynsberghe (2019), Umbrello and Van de Poel (2021), Umbrello et al. (2021)
	T2	Designing systems to support the identified values or supporting/hindering values by existing technological properties	Miller et al. (2007), Betz and Fritsch (2016), Yoo et al. (2013), Xu et al. (2012), Umbrello and De Bellis (2018), Xu et al. (2008), Badillo-Urquiola et al. (2020), Borning et al. (2004), Vermaas et al. (2010), Royakkers and Steen (2017), Friedman et al. (2006), Cawthorne and Robbins-van Wynsberghe (2019), Barn and Barn (2018), Friedman et al. (2002), Borning et al. (2005), Cummings (2006), Cheon and Su (2016), Ghosh (2016), Friedman and Borning (2002), Thornton et al. (2018), Heger (2019), Cawthorne and Cenci (2019), Harbers and Neerincx (2017), Walton and DeRenzi (2009), Detweiler and Hindriks (2012), Van Andel et al. (2015), Vernim et al. (2022), Oktay and Pender (2020), Boyd (2022), Helbing et al. (2021), Jaljolie et al. (2022), Sapraz and Han (2021), Köhler et al. (2022), Cenci and Cawthorne (2020), Kowe et al. (2022), Cruz-Martínez et al. (2021), Iversen et al. (2020), Graubohm et al. (2020), Umbrello et al. (2021), Primiero et al. (2020)
	T3	Evaluating the design, adoption, and use of the designed systems	Miller et al. (2007), Xu et al. (2012), Friedman et al. (2002), Borning et al. (2005), Harbers and Neerincx (2017), Vernim et al. (2022), Lüthi et al. (2021), Van Wynsberghe (2012), Boyd (2022), Wambsganss et al. (2021), Iqbal et al. (2021), Primiero et al. (2020)

Table 8
Coverage of the primary studies with respect to the conducted activities to embed the (ethical) values in design and development of software systems.

Studies	C1	C2	C3	C4	C5	E1	E2	E3	T1	T2	T3
Alsheikh et al. (2011), Maathuis et al. (2019), Detweiler et al. (2010a)	■	■	■	□	■	■	□	□	□	□	□
Miller et al. (2007)	■	■	■	□	■	■	□	■	□	■	■
Betz and Fritsch (2016)	■	■	■	□	■	□	□	■	□	■	□
Alvarenga and Tanev (2017), Van Wynsberghe (2013), Dadgar and Joshi (2015b), Le Dantec and Poole (2008), Grünloh (2018), Harbers et al. (2010)	■	■	■	□	□	□	□	□	□	□	□
Walldius et al. (2005, 2015)	■	■	■	■	□	□	□	□	□	□	□
Yoo et al. (2013), Royakkers and Steen (2017), Barn and Barn (2018), Cheon and Su (2016), Ghosh (2016), Sapraz and Han (2021), Kowe et al. (2022), Cruz-Martínez et al. (2021)	■	■	■	□	□	■	□	□	□	■	□
Nouwen et al. (2015), Zaman and JafariNaimi (2015), Nouwen and Zaman (2018)	■	■	■	□	■	■	□	■	□	□	□
Xu et al. (2012), Harbers and Neerincx (2017), Boyd (2022)	■	■	■	□	□	■	□	□	□	■	■
Detweiler et al. (2010b), Le Dantec et al. (2009), Briggs and Thomas (2015), Sambasivan and Jackson (2008), Denning et al. (2014), Dadgar and Joshi (2015a), van de Kaa et al. (2020), Mithun et al. (2018), Burrows et al. (2015), Harbers and Neerincx (2014), Kumari et al. (2021), Smits et al. (2022), García and Gil (2020)	■	■	■	□	□	■	□	□	□	□	□
Friedman and Kahn (2000)	□	■	■	□	□	□	□	□	□	□	□
Umbrello and De Bellis (2018), Thornton et al. (2018), Cawthorne and Cenci (2019)	■	■	■	□	■	□	□	□	□	■	□
Xu et al. (2008), Vermaas et al. (2010), Friedman et al. (2006), Cummings (2006), Walton and DeRenzi (2009), Detweiler and Hindriks (2012), Cenci and Cawthorne (2020), Iversen et al. (2020)	■	■	■	□	□	□	□	□	□	■	□
Simons and Verhagen (2008), Sackey (2020), Watkins (2018), Rychwalska and Roszczynska-Kurasinska (2017), Umbrello and Yampolskiy (2022)	■	■	■	□	■	□	□	□	□	□	□
Umbrello (2019)	■	■	■	□	■	□	□	□	■	□	□
Badillo-Urquiola et al. (2020), Heger (2019), Jaljolie et al. (2022), Köhler et al. (2022)	■	■	■	□	■	■	□	□	□	■	□
Borning et al. (2004), Friedman and Borning (2002)	■	■	■	■	□	□	□	□	□	■	□
Cawthorne and Robbins-van Wynsberghe (2019)	■	■	■	□	□	□	□	□	■	■	□
Friedman et al. (2002), Vernim et al. (2022)	■	■	■	□	□	□	□	□	□	■	■
Borning et al. (2005)	■	■	■	■	□	■	□	□	□	■	■
Ast and Gaedke (2016)	□	■	■	■	□	□	□	□	□	□	□
Van Andel et al. (2015)	■	■	■	□	□	■	■	□	□	■	□
Gan and Moussawi (2022), Burmeister et al. (2021)	□	□	■	□	□	□	□	□	□	□	□
Lüthi et al. (2021)	■	■	■	□	□	■	□	■	□	□	■
Oktay and Pender (2020)	□	□	□	□	□	■	□	□	□	■	□
Van Wynsberghe (2012)	□	□	■	□	□	□	□	□	□	□	■
Wambsganss et al. (2021)	□	□	■	□	□	■	□	□	□	□	■
Helbing et al. (2021)	■	□	■	□	□	□	□	□	□	■	□
Umbrello and Van de Poel (2021)	■	■	■	□	□	□	□	□	■	□	□
Umbrello (2020)	■	□	■	□	■	□	□	□	□	□	□
Iqbal et al. (2021)	□	□	□	□	□	□	□	□	□	□	■
Tokranova (2020)	■	□	■	□	□	□	■	□	□	□	□
Dexe et al. (2020)	■	□	■	□	□	■	□	□	□	□	□
Graubohm et al. (2020)	■	□	■	□	■	■	□	□	□	■	□
Umbrello et al. (2021)	□	□	■	□	□	□	□	□	■	■	□
Primiero et al. (2020)	□	□	■	□	□	■	□	□	■	■	□

4.4.1. Representing ethical values as software requirements

As values are intrinsically ambiguous and supporting them in software design process is non-trivial, there is a need to translate values into tangible constructs (like software requirements) and make them concrete and operational. For example, in Maathuis et al. (2019), the users of mental health (mHealth) apps must be confident and have trust in mental health care. However, the ambiguous nature of trust in mHealth forces developers to translate this value into requirements that foster and make it concrete, such as security, privacy, and good access to information and communication technologies infrastructure.

Translation of values into software requirements. Translation of values into software requirements is a relatively neglected aspect in software design, it was conducted in 4 out of 85 studies (*i.e.,* Umbrello (2019), Cawthorne and Robbins-van Wynsberghe (2019), Umbrello and Van de Poel (2021), Umbrello et al. (2021)), using the method of the *value hierarchy*. By translating values into more concrete and tangible constructs, this method ensures that the design sufficiently reflects the ethical values.

A value hierarchy is a structure that is proposed by VSD methods and consists of three layers: (i) values, (ii) norms, and (iii) software requirements. For example, consider using this structure in the context of blood sample transportation drones (cf. Cawthorne and Robbins-van Wynsberghe (2019)). The *values layer* (*i.e.,* upper layer) consists of values that are independent from the context in which the system is situated, *e.g., safety*. The *norms layer* (*i.e.,* middle layer) consists of norms that are context-dependent in contrast to values, and could be any prescription or restriction for action. In this example, this layer includes the norm relevant to safety, *e.g., using drones is safer than driving with respect to fatalities per round-trip*. The *software requirements layer* (*i.e.,* lower layer) consists of software requirements that are obtained by further investigation of the norms. In this example, this layer includes the requirements: *minimizing weight to less than 1.5* kg, *having components with less than 300* g, and *minimizing impact speed to less than 60* km/h.

5. Discussion

In this section, we discuss (per research question) the implications of our research findings. Furthermore, we outline the future research directions that emerge from our study.

5.1. Identification of key stakeholders (RQ1)

In Section 4.2, we proposed a classification of the stakeholder roles in terms of receiving benefit or harm from the system. This classification encompasses stakeholders who interact directly with the system (*i.e.,* use or build the system) and passively involved stakeholders who are indirectly affected by the system.

Indirect stakeholders are not often identifiable in software design. Our primary studies identified two categories of stakeholders: (i) those who interact directly with the system (*i.e., visible stakeholders*[17]) and (ii) those who do not directly interact with the system but may be affected by it (*i.e., invisible stakeholders*[18]). The findings presented in Table 2 highlight a significant emphasis on direct stakeholders, including both system users and system development organization, as identified in 74 and 38 out of 85 studies, respectively. In contrast, only 27 out of 85 studies focused on indirect stakeholders. Drawing from these findings, we conjecture that identifying direct stakeholders in software design is well recognized, while indirect stakeholders are often ignored as they depend on the different contexts in which the system is intended to be used. These stakeholders remain invisible to individuals involved in software development. To illustrate this point, consider the case of blood sample transportation drones (Cawthorne and Cenci, 2019), unmanned aerial vehicles for swiftly transporting medical supplies, especially blood samples, between healthcare facilities. These drones show potential in healthcare by enhancing efficiency, even in challenging terrain or during humanitarian crises. In this particular case, specific stakeholders, including medical personnel, patients, and drone manufacturers, are easily identifiable during the system design process. However, stakeholders like local communities and society are invisible and are often overlooked as they do not directly interact with the system.

Whilst the identification of the indirect stakeholders is crucial, it is not trivial due to the following three reasons: (i) existing methods (such as VSD) do not provide any systematic way to identify invisible stakeholders or the far-reaching impacts of systems. (ii) Conventional scoping activities in software engineering primarily focus on identifying stakeholders that directly interact with the system, *e.g.,* through use cases (de Moraes et al., 2009). Indirect stakeholders, the ones that do not interact with the system, are deemed out of scope. Naturally, the resulting system does not account for its effects on these stakeholders who are invisible. (iii) Anticipating who might be indirectly affected by the system requires anticipating human–system interactions as well as the contexts in which those interactions occur, *beyond* the immediate scope of the system and in a creative manner. As such, there is a need to develop ways to identify indirect stakeholders and their concerns.

Given the above-mentioned reasons, we distinguish the following research directions: (i) develop methods to anticipate the overarching effects of the system on individuals and society, (ii) develop techniques to explicitly include indirect stakeholders into the scope of the system, and (iii) explore systematic and creative ways to anticipate human–system interactions and use contexts of the system.

Lack of focus on the concerns of indirect stakeholders. As shown in Table 7 (in online Appendix D), we found that studies only focus on discovering the concerns of direct stakeholders, *i.e.,* system users and system development organization. However, there is a notable lack of emphasis on identifying the concerns of indirect stakeholders. This implies a limited focus on the identification of invisible stakeholders and their concerns. This limited focus emphasizes the need for broader attention and research towards identifying the concerns of these stakeholders.

Not making explicit the concerns of such stakeholders during software design implies ignoring potential ethical issues and harms that systems might cause to them. In the example of blood sample transportation drones (Cawthorne and Cenci, 2019), ignoring the concerns of local communities and society during the design process could lead to building a system that may endanger the individuals' physical welfare. This neglect could also result in risks, such as increasing the likelihood of drones colliding with aircraft or causing harm to individuals on the ground.

To address the consequences of such neglect, we advocate for a research direction centered on the development of methods like the *social implication design method*[19] (Tromp and Hekkert, 2014). The methods that have the potential to assist designers in proactively addressing potential social issues and incorporating the concerns of invisible stakeholders when designing products or services.

Stakeholder classification can help identify direct and indirect stakeholders. In general, an explicit stakeholder classification helps in identifying the relevant stakeholders of a software system (Preiss and Wegmann, 2001; Bedjeti et al., 2017). In particular, for ethics, however, our analysis of the primary studies revealed that such classification is

[17] They usually can be identified by the stakeholder analysis methods when designing systems.

[18] In contrast to visible stakeholders, they are difficult to be identified by existing stakeholder identification methods when designing systems.

[19] This method focuses on the hidden influence of design on individuals, particularly those who might not be readily visible as stakeholders, and the consequential social implications.

lacking. We conjecture that our proposed classification for stakeholder roles (see Fig. 5 and Table 2) can be a step towards filling this gap. This classification divides the stakeholders into three roles based on how they interact with the system (*i.e.,* system users, system development organization, and indirect stakeholders). By making explicit both direct and indirect stakeholders, it enables design decision makers to consider a wider range of stakeholders and be cognisant of indirect stakeholders that would otherwise become invisible.

5.2. Identification of ethical values (RQ2)

When developing software systems, it is essential to recognize that different stakeholders hold distinct views and values regarding the system. Software engineers often prioritize ethical values tied to system implementation, product managers may align their values with organizational profits, and end users typically focus on values associated with meeting their needs and achieving their goals. This diversity of perspectives has significant implications for identifying ethical values within software systems. In the following, we discuss these implications in more detail.

Stakeholders' concerns provide context to discover ethical values. Concerns of stakeholders in a specific context are related to perceived risks and issues that can potentially affect their values. Based on the primary studies, we found that the identification of stakeholders' concerns could provide contexts to discover the values at stake. Discovering the links between concerns and values could help take ethical values into account as well as uncover conflicting concerns and values for design space exploration. As an illustration, consider parental control applications in which parents have concerns about the safety risks of teens (Nouwen and Zaman, 2018). In such cases, concerned parents may discover different *safety* and *security* issues and potential conflicts between *safety* and *involvement*.

Also, as shown in Table 7 (in online Appendix D), different stakeholder concerns might correspond to common values. In such cases, however, although the types of values may be the same for different stakeholders, they may differ with respect to the specific interests of each stakeholder role. For example, privacy violation can have different meanings for end users (*e.g.,* protect access to identity data of young children) and system development organisation (*e.g.,* comply with the "integrity and confidentiality" principle of the GDPR). To satisfy this value for each role, we may need to make different ethics-related decisions and select different design options.

Methods and techniques for value extraction. During our examination of value elicitation from system stakeholders, we found different methods and techniques (see Table 5), each suited for specific purposes and situational contexts. *Workshops*, for instance, facilitate collaborative analysis of values-related issues and ethical values, while *sketches* provide a non-verbal understanding of the elicited values. Despite the usefulness of each method and technique in extracting values from stakeholders, there remains a significant lack of understanding regarding their nature and appropriateness.

Methods and techniques employed for requirement elicitation can be tailored and developed for value elicitation (Alidoosti et al., 2022). The abstractness and ambiguities inherent in ethical values make dealing with values more intricate and challenging than addressing requirements, particularly in diverse ethical situations. Consequently, there arises a need to employ specific methods and techniques that align with the nature of values when extracting them from stakeholders. Addressing this gap prompts a valuable future direction: the enhancement of existing requirements engineering methodologies and techniques to account for the distinctive characteristics of norms and values. Such adjustments would enable the precise extraction of values directly from stakeholders.

The value model helps define ethical values and recognize relations among them. Our value classification is a first attempt to model values systematically, and it is a prerequisite for explaining and interpreting software ethical values. We proposed a value model based on the Schwartz value structure (see Fig. 2) for value classification. This model provides an opportunity to define software ethical values according to the value categories and their motivational goals.

Also, the circular arrangement of the model could help recognize the relations among the values. More precisely, the location of a specific value in the model could help discover the relation of that value with other values. Accordingly, the closer any two values are, the more similarity they have in terms of underlying motivations and vice versa. For example, in Fig. 2, consider the value of *flexibility* in the dimension of *conservation*. According to the value model, it can be recognized that the values located in *openness to change* are likely to have a contradictory relation with it, and the values located in its adjacent dimension (*i.e., self-transcendence vs. self-enhancement*) can be congruent with it.

Insufficient focus on certain ethical values. Analyzing Table 3, it becomes evident that values such as *privacy*, *trust*, *autonomy*, and *welfare*, have received considerable attention in existing literature, with 40, 40, 35, and 29 studies dedicated to each, respectively. However, certain values like *social order* and *accuracy* have received comparatively little exploration and investigation, each being mentioned in only one paper. Neglecting these unexplored values in software design can lead to unforeseen ethical risks and harms for stakeholders, impeding effective risk mitigation and potentially endangering lives.

The value model resulting from our study could help in raising awareness among design decision makers about the existence of unexplored ethical values and in attracting research that will address them whenever risky to do otherwise.

As described in Section 2.1.2, this value model comprises ten value categories that depict the relationships among these categories. To leverage this model, software designers can follow a structured approach. Initially, they can pinpoint the value categories that align with the system's goal(s). Next, they can identify the sub-values in relation to stakeholders to determine if the system supports or compromises them, and discern the relationships between these values, including any conflicts or congruencies.

Insufficient focus on value relations.

Table 6 highlights a predominant focus in primary studies on conflicts and tensions among values, overlooking the potential for values to reinforce one another and the existence of congruity between them. Neglecting value relations in software design can undermine ethical decision making, heighten ethical issues, and potentially result in unintended negative consequences for individuals and society.

As discussed in Section 4.3.3, Fig. 6 serves as an initial step towards recognizing potential value relations. This research seeks to identify, analyze, and contextualize value relations, including conflicts and tensions, with the aim of enhancing ethical decision making in software design. Further empirical studies of value relations to explore value relations in diverse ethical situations during the software design process can shed more lights.

Inadequacy in value conflict resolution. When designing for a range of values (instead of a single one), it is possible that a design option that supports one value would compromise another value. This necessitates the consideration of trade-offs among ethical values, which can be revealed through the exploration of different design options.

We found only a few primary studies (Miller et al., 2007; Betz and Fritsch, 2016; Nouwen et al., 2015; Zaman and JafariNaimi, 2015; Nouwen and Zaman, 2018; Lüthi et al., 2021) that propose solutions to resolve conflicts among the values in the software design process. While the software engineering field has encountered this challenge, there are valuable methods from other domains that can serve as sources of inspiration. For instance, methods such as "value alignment" and "value dams and value flows" from various other fields provide valuable insights that can enhance approaches in software engineering. These methods can be instrumental in effectively managing and resolving value conflicts within software design.

Based on the limited research on value conflict resolution, a promising research direction centers on understanding how value conflicts can be dealt with and, hence, embedded in methods and techniques. Furthermore, this research could explore the feasibility and selection of qualitative and quantitative techniques for facilitating trade-offs among values and their associated possible design options. Delving deeper into these areas can advance our understanding and develop practical methods for managing value conflicts during the design process.

5.3. Embedding requirements and values (RQ3)

To integrate the requirements and values of stakeholders, software researchers and developers must revisit and create software engineering methods and techniques with respect to ethical considerations tailored to software design and development.

Varying support for VSD.
Table 7 reveals that VSD stands out as the sole method used to address ethical values during the software design process. However, the primary studies cover only a proper subset of the VSD activities, particularly those pertaining to the conceptual ones addressing design (see Table 8).

This limited coverage of VSD activities can be attributed to either (i) the narrow focus of the studies themselves, or (ii) the inherent challenge of conducting the complete range of VSD activities to address ethical considerations.

As an example of the studies with a narrow focus, Detweiler and Hindriks (2012) emphasizes the use of design patterns to help designers incorporate value considerations in the design process. However, the study does not delve into resolving value conflicts. In this example, the focus is deliberately limited to conceptual and design-related VSD activities, specifically $C1$ (*i.e.,* identifying stakeholders), $C2$ (*i.e.,* identifying benefits and harms imposed by the system on stakeholder groups), $C3$ (*i.e.,* mapping benefits and harms onto corresponding values), and $T2$ (*i.e.,* designing systems to support the identified values or supporting/hindering values by existing technological properties).

As an example highlighting challenges in incorporating ethical considerations, Zaman and JafariNaimi (2015) discusses the difficulties in identifying and applying human values in design, *e.g.,* finding suitable ways to facilitate discussions about values among different stakeholders or accounting for the voices of stakeholders while considering their values with respect to their power in the organization. In light of the aforementioned challenges, certain VSD activities, namely $C4$ (*i.e.,* categorizing the identified values), $E2$ (*i.e.,* prioritizing the values), $T1$ (*i.e.,* translating the identified values into software requirements), $T2$ (*i.e.,* designing systems to support the identified values or supporting/hindering values by existing technological properties), and $T3$ (*i.e.,* evaluating the design, adoption, and use of the designed systems), are excluded from consideration.

Insufficient focus on operationalizing ethical values in software design. Embedding intrinsically ambiguous ethical values into software systems requires values to be made as concrete as possible for design decision making.

We found only four primary studies (Umbrello, 2019; Cawthorne and Robbins-van Wynsberghe, 2019; Umbrello and Van de Poel, 2021; Umbrello et al., 2021) discussing the translation of values into software requirements. The process of translating values into software requirements essentially involves the conversion of ethical values into specific and concrete criteria or specifications that can guide the development of software systems. In essence, it is about defining and articulating how these values should be integrated into the practical aspects of software design and development. This limited number of studies signifies a lack of attention given to converting values into tangible software requirements, which is essential for supporting values in software design. To address this gap, a research direction involves investigating methods to make values more tangible and to facilitate their translation into goals and software requirements. This endeavor aims to facilitate the practical implementation of ethical values in software design and development.

6. Threats to validity

In this section, we discuss the main threats that may have affected the validity of our study. We organize them according to the classification by Wohlin et al. Wohlin et al. (2012) and by using the considerations by Lago et al. Lago et al. (2024).

Construct validity refers to the correct data collection and the correct measurement of the theoretical concepts (Easterbrook et al., 2008).

- One typical threat in SLRs can be related to selecting unsuitable search engines. To mitigate this threat, Google Scholar was used as a tool to collect papers (as explained in Section 3.2) since it is a meta-engine and is commonly used in SLRs. In addition, Martín-Martín et al. (2018a) provide empirical evidence that Google Scholar citation data is essentially a super-set of other online bibliographic databases.
- Not including studies that focus on relevant sub-fields (like AI) when studying SE ethics may introduce a threat to this validity. We consider that this threat was mitigated by the fact that the potential issues found in this study are similar to those in relevant sub-fields. In this way, our proposed solutions (like a value model) can be used for dealing with such issues (*e.g.,* identifying value relations), too.
- Ill-defined search queries can result in a large number of irrelevant studies. In order to mitigate this threat, we defined two search queries to ensure covering the potentially relevant studies (as discussed in Section 3.2). By using the first query, we covered a feasible number of studies, but there was a risk of overlooking the studies containing the term “value”. So, to cover the studies including this term and the relevant phrases like “value-sensitive” and “value-centered” that focus on SE ethics, we defined another query (*i.e.,* search query 2).
- Choosing an improper classification for the values can introduce some bias to the analysis. We have mitigated this threat by selecting the Schwartz value structure, a widely adopted classification scheme in social sciences. This value classification was led to reducing researchers’ bias since there was no need to develop our own classification.

Internal validity aims to ensure that the collected data enables researchers to draw valid conclusions (Creswell and Creswell, 2017).

- One typical threat to internal validity in SLRs is bias in extracting enough data from the studies, influencing the accuracy of the extracted data and the result. To mitigate this threat, we extracted also descriptive data concerning our research questions.
- There is a threat to internal validity due to improperly defining and conceptualizing the extracted values from the studies. This threat might result from the experience level of researchers in relation to philosophy and social science for value conceptualization. In our study, this threat was mitigated by involving two researchers in defining values, and in the presence of disagreements, by consulting the other researchers.
- Classifying values can introduce some bias to the analysis as values are intrinsically ambiguous and can be related to more than one value category. To mitigate this threat and facilitate the value classification, we took advantage of the presented meta-inventory of human values in Cheng and Fleischmann (2010). Also, we involved three researchers in providing an objective classification.

7. Conclusion

Software systems are ubiquitously employed in society and increasingly integrated into all aspects of people’s lives. The proper functioning of software systems in both technical and ethical dimensions is essential, as individuals and society at large can be affected by these systems in many facets, from healthcare to communication and education to economic welfare. These systems can affect individuals by undermining their values and causing ethical issues, such as causing harm to the environment, violating privacy, and discriminating amongst people.

This article outlines a study on the application of ethics in software design and development. It highlights the focus on stakeholder values and their ethical implications based on a systematic literature review (SLR). By analyzing 85 primary studies, three key stakeholder roles were identified: system users, the system development organization, and indirect stakeholders, all from an ethical value perspective.

This study categorized ethical values commonly used in software design according to the Schwartz value structure–a tool to help software engineers conceptualize values, recognize value relations, and identify potential conflicts during design and development. Our value model can be treated as a general categorization within the broader field of software engineering. Relevant subfields, such as value-driven requirements engineering (Thew and Sutcliffe, 2018) or the ethics of AI-enabled systems (Vakkuri et al., 2021; Siqueira De Cerqueira et al., 2021; Leikas et al., 2019; Halme et al., 2022), can treat this generic model as a foundation, customizing and extending it to fit their specific needs, which could be the focus of future studies .

Our analysis has revealed challenges relating to the identification of indirect stakeholders and their hidden concerns in software design, their concerns are omitted because they are invisible to developers. We classify stakeholders into different groups to raise awareness of hidden stakeholders. We propose an enhanced Schwartz value structure which can be useful for identifying and structuring ethical values for software development. Based on the values identified in the primary studies, we relate value conflicts and tensions to understand their relationships and to facilitate requirement exploration. We also found methods and challenges for value identification and requirement translation. This finding prompts a call to enhance value elicitation to integrate ethical values into software design.

The findings suggest several promising future research directions: (i) providing a systematic way to identify relevant stakeholders (and especially invisible stakeholders) in software design, (ii) proposing methods and techniques to elicit ethical values from system stakeholders that suit different ethical purposes and situations, (iii) providing a comprehensive list of ethical values used in software design along with their definitions, (iv) proposing a model to recognize relations among values during the design process, (v) proposing suitable methods to resolve value conflicts, and (vi) developing methods to operationalize ethical values in software design.

Understanding existing issues in SE ethics and having a model for classifying ethical values provide insights into other related areas, such as AI. We hope that further work in this area can inspire software engineers to consider ethical values as important requirements in software design and development.

CRediT authorship contribution statement

Razieh Alidoosti: Conceptualization, Data curation, Formal analysis, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Patricia Lago:** Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Maryam Razavian:** Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Antony Tang:** Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jss.2025.112430.

Data availability

Data will be made available on request.

List of primary studies

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Yücel, R., Elibol, H., Dağdelen, O., 2009. Globalization and international marketing ethics problems. Int. Res. J. Financ. Econ. 26, 93–104.

Razieh Alidoosti is a Ph.D. student at the Computer Science Department of Vrije Universiteit Amsterdam and Gran Sasso Science Institute in Italy. Her research interests include software architecture and ethics in software engineering.

Patricia Lago is a full professor of software engineering at Vrije Universiteit Amsterdam. She received her Ph.D. in Control and Computer Engineering from Politecnico di Torino. She is the recipient of an Honorary Doctorate at the Norwegian University of Science and Technology (NTNU) for her scientific contribution to software sustainability. Her research is at the intersection of software architecture, sustainability, and quality assessment. She is a member of ACM and IEEE.

Maryam Razavian is an assistant professor of industrial engineering at Eindhoven University of Technology. Her research interests include software design reasoning, human aspects of software design, software architecture, and service orientation. She received her Ph.D. in Computer Science from Vrije Universiteit Amsterdam. She is a member of ACM and IEEE.

Antony Tang is an associate professor at Swinburne University of Technology. His research interests include software architecture design reasoning, software development processes, and software architecture knowledge management. He received his Ph.D. in Information Technology from Swinburne. Prior to being a researcher, he had spent many years designing and developing software systems.